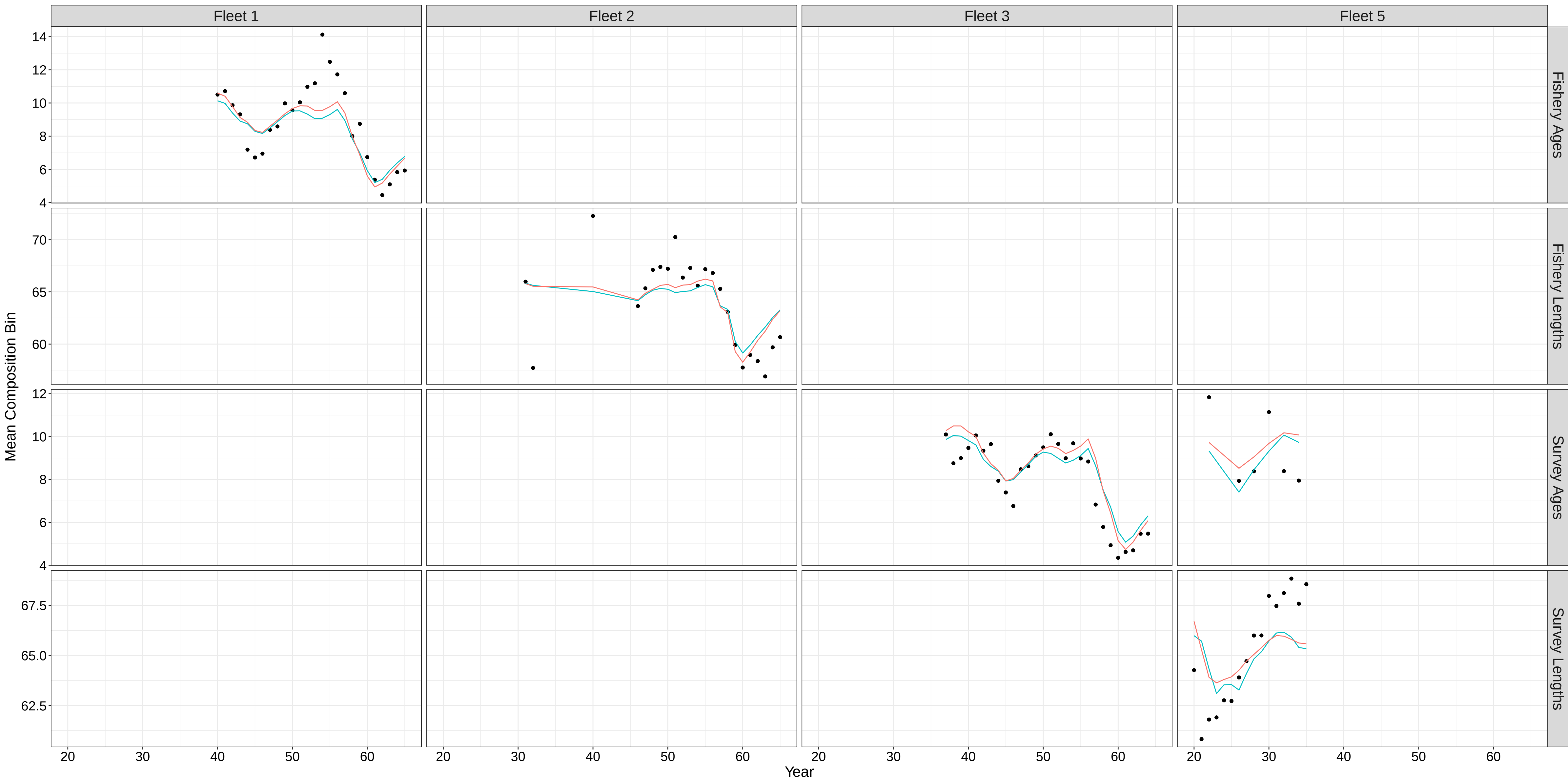


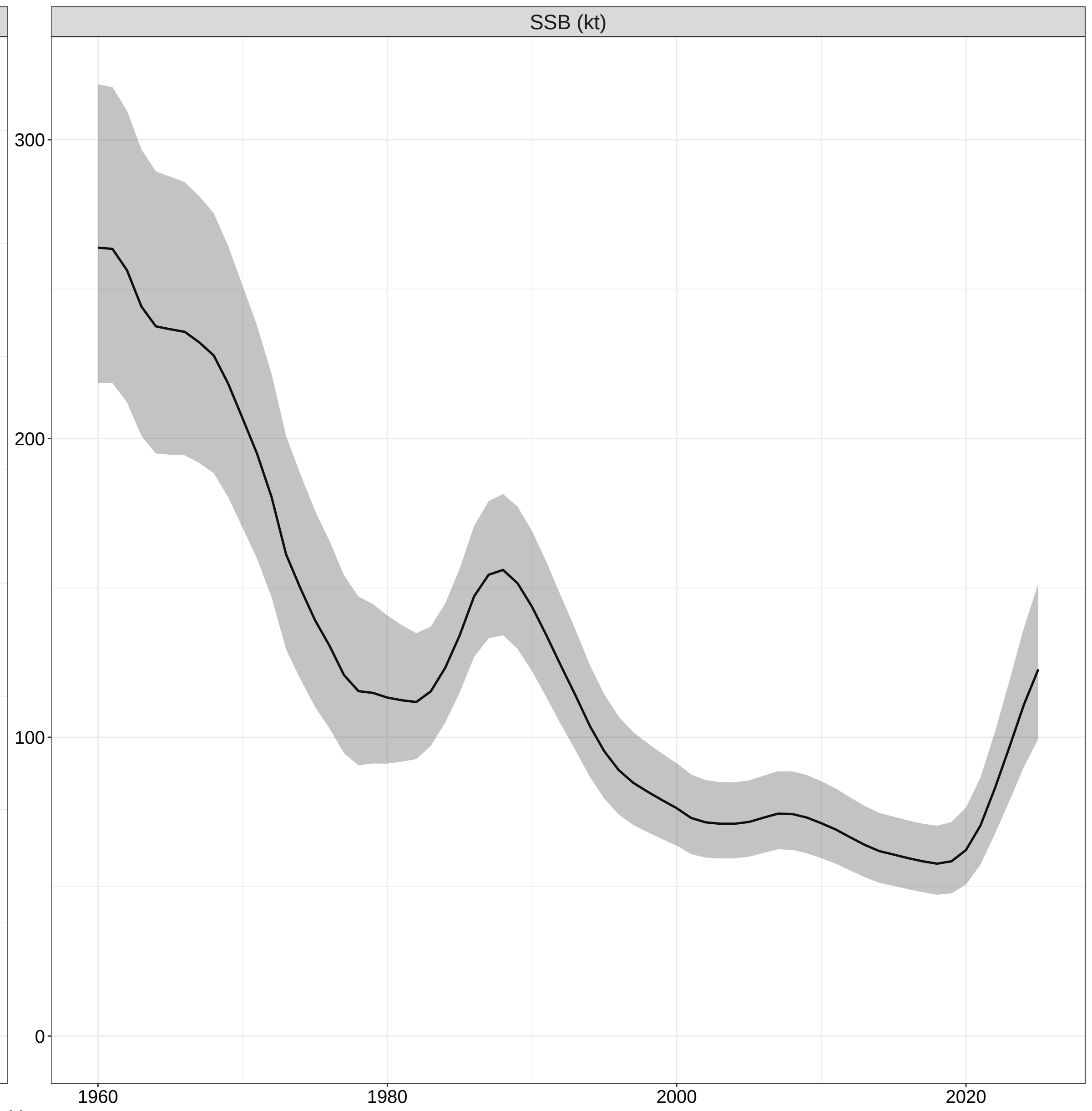
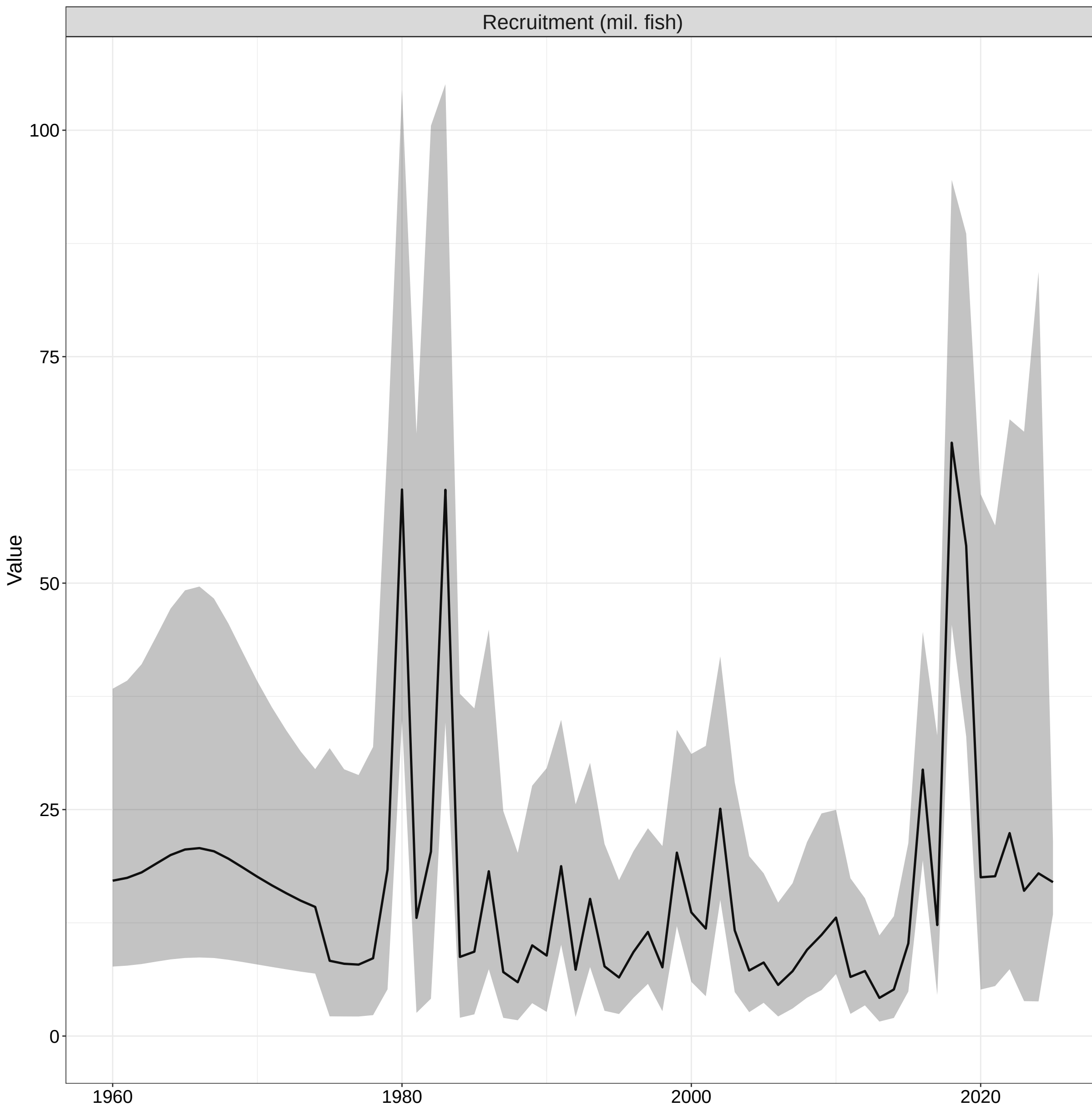
Source	SE	Eff_SE	Data_Wt	ISS	ESS
BS LL Survey Index	0.20	0.20	1.00	NA	NA
AI LL Survey Index	0.24	0.24	1.00	NA	NA
GOA LL Survey Index	0.08	0.08	1.00	NA	NA
JPN LL Survey Index	0.07	0.07	1.00	NA	NA
Fixed Gear Fish Ages	NA	NA	0.10	145.15	14.15
BS LL Survey Ages	NA	NA	NA	NA	NA
AI LL Survey Ages	NA	NA	NA	NA	NA
GOA LL Survey Ages	NA	NA	0.30	171.43	51.99
JPN LL Survey Ages	NA	NA	0.16	83.33	12.96
Trawl Fish Lengths	NA	NA	0.23	45.39	10.24
JPN LL Survey Lengths	NA	NA	0.36	54.62	19.72

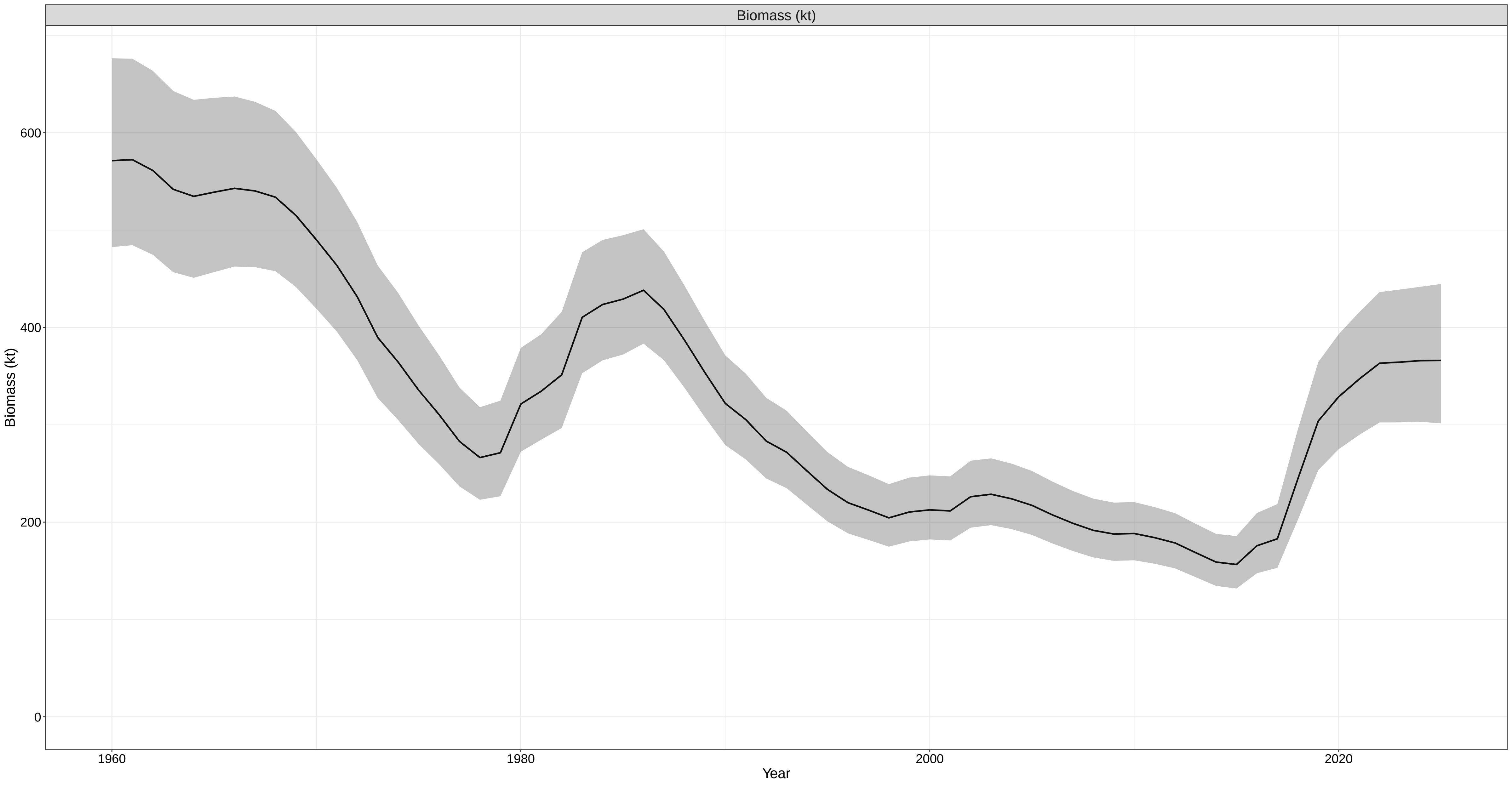
G3;Successfully passed post-optim-sanity checks

colour — First Iter — Last Iter



	Model	Region	Terminal_SSB	Terminal_SSB0	Terminal_F	Catch_Advice	B_Ref_Pt	F_Ref_Pt	B_over_B_Ref	B_over_DynB_Ref	F_over_F_Ref
1	1	1	122.76257	283.32105	0.0820013975298393	21.86816	114.94984	0.07507	1.06797	0.4333	1.09231

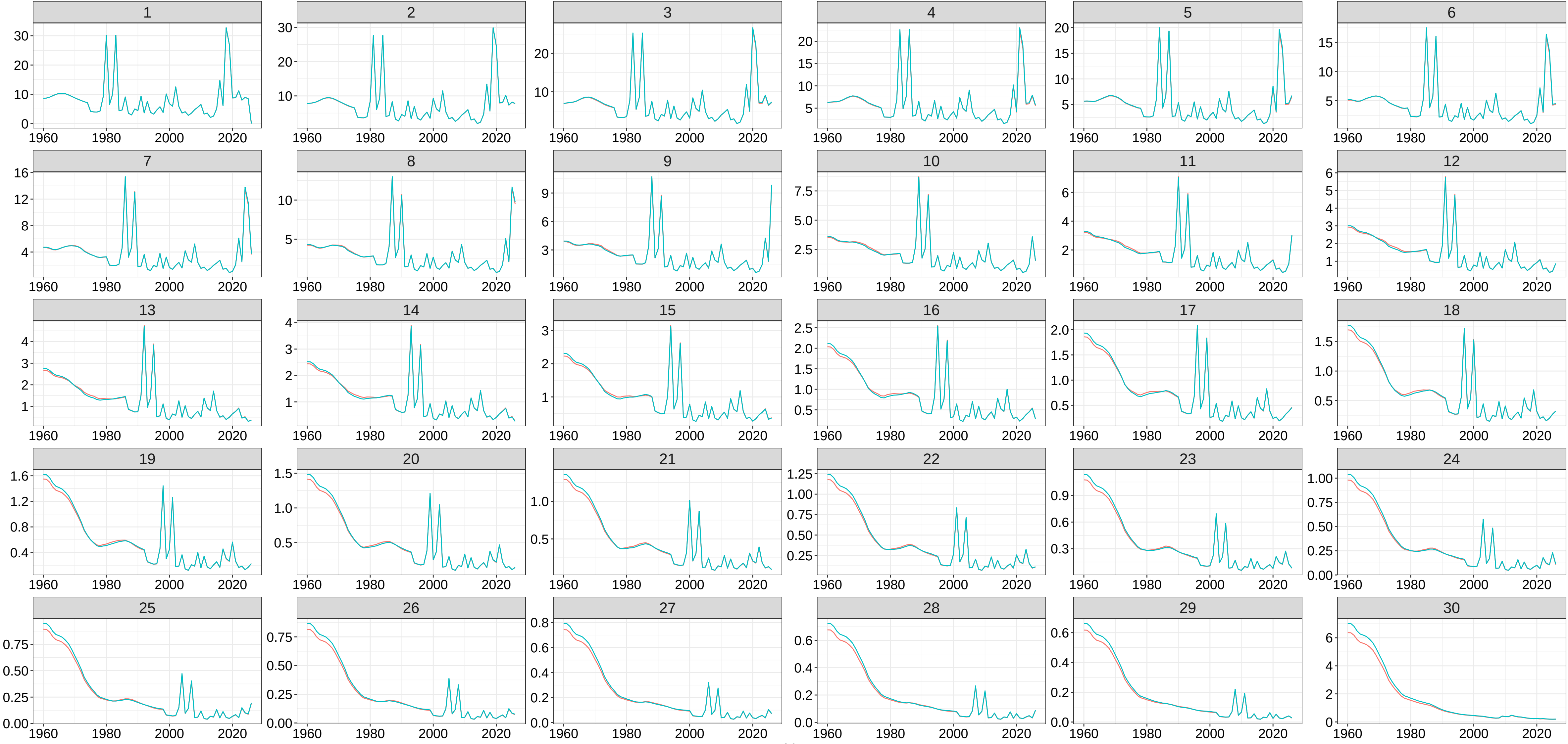




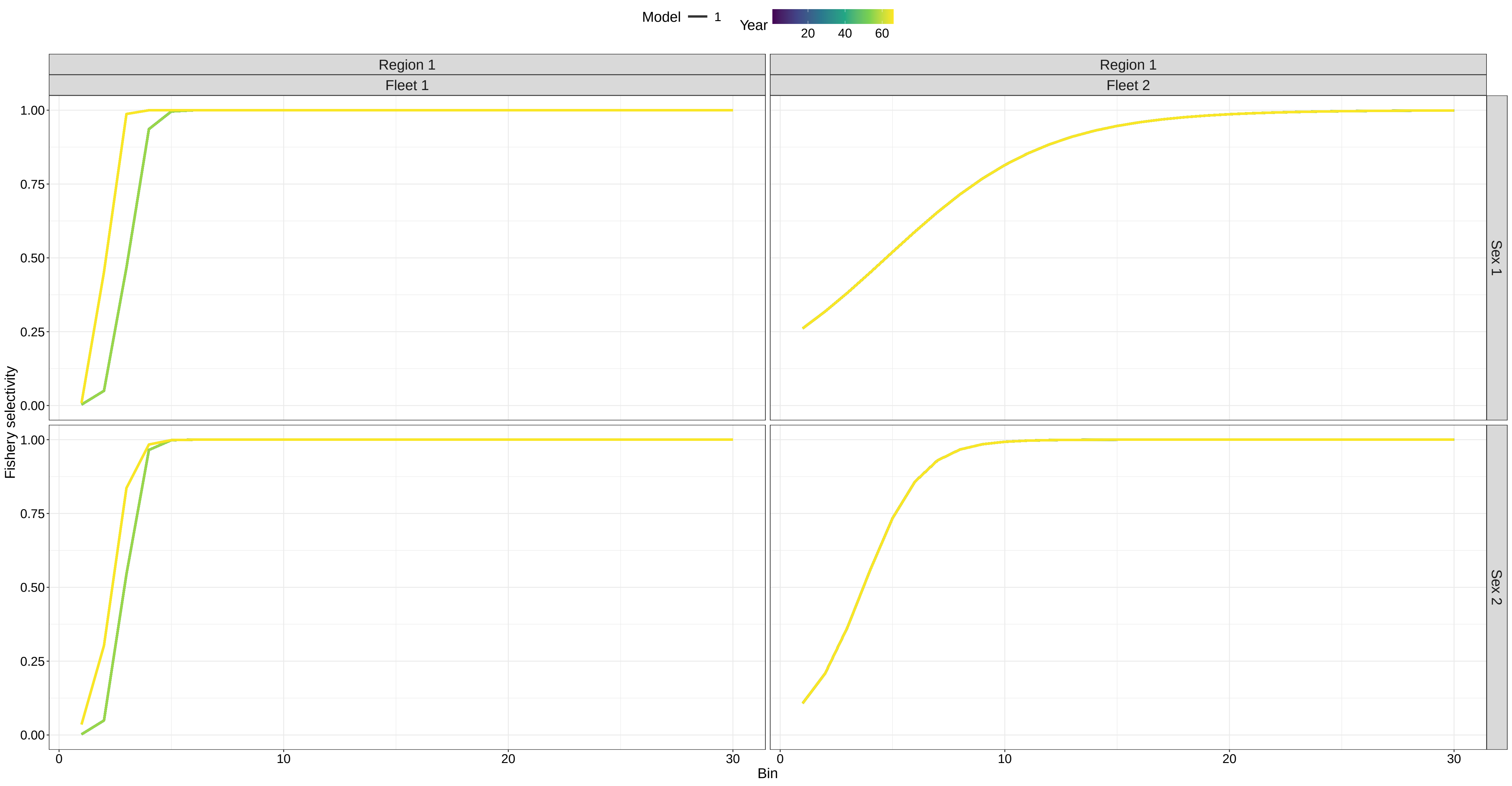
Numbers-at-Age

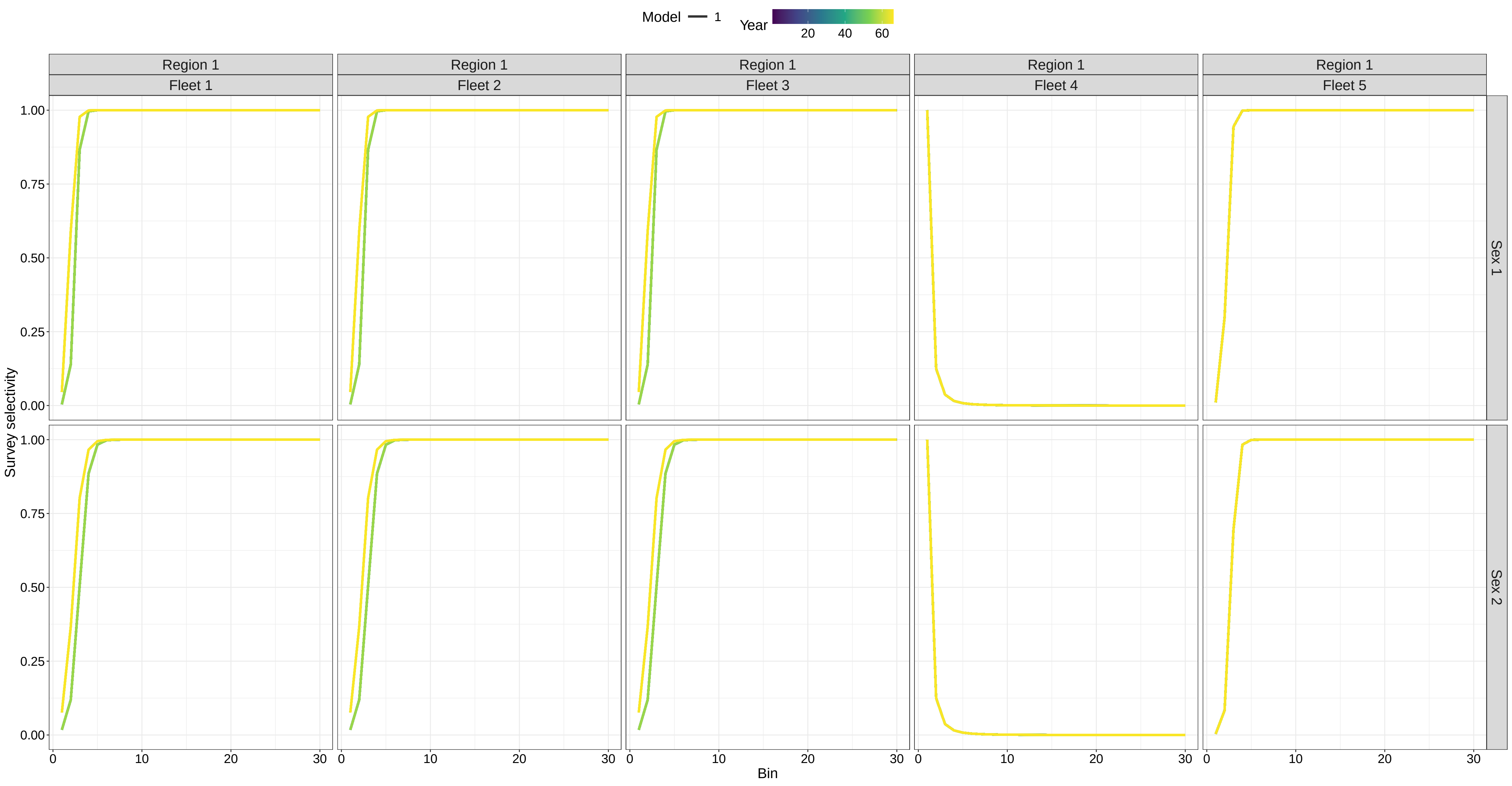
Sex — Female — Male

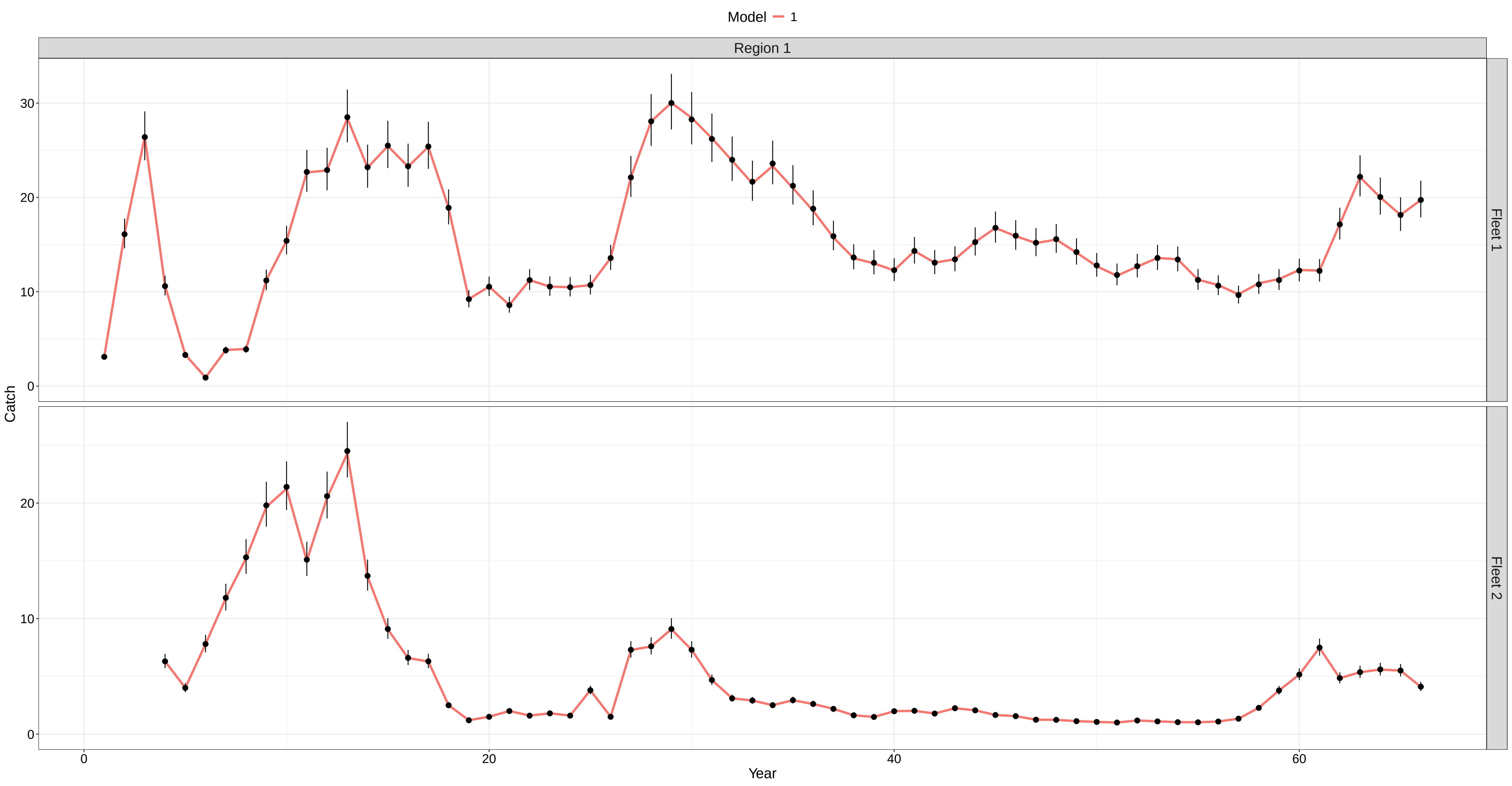
Numbers-at-age (millions)



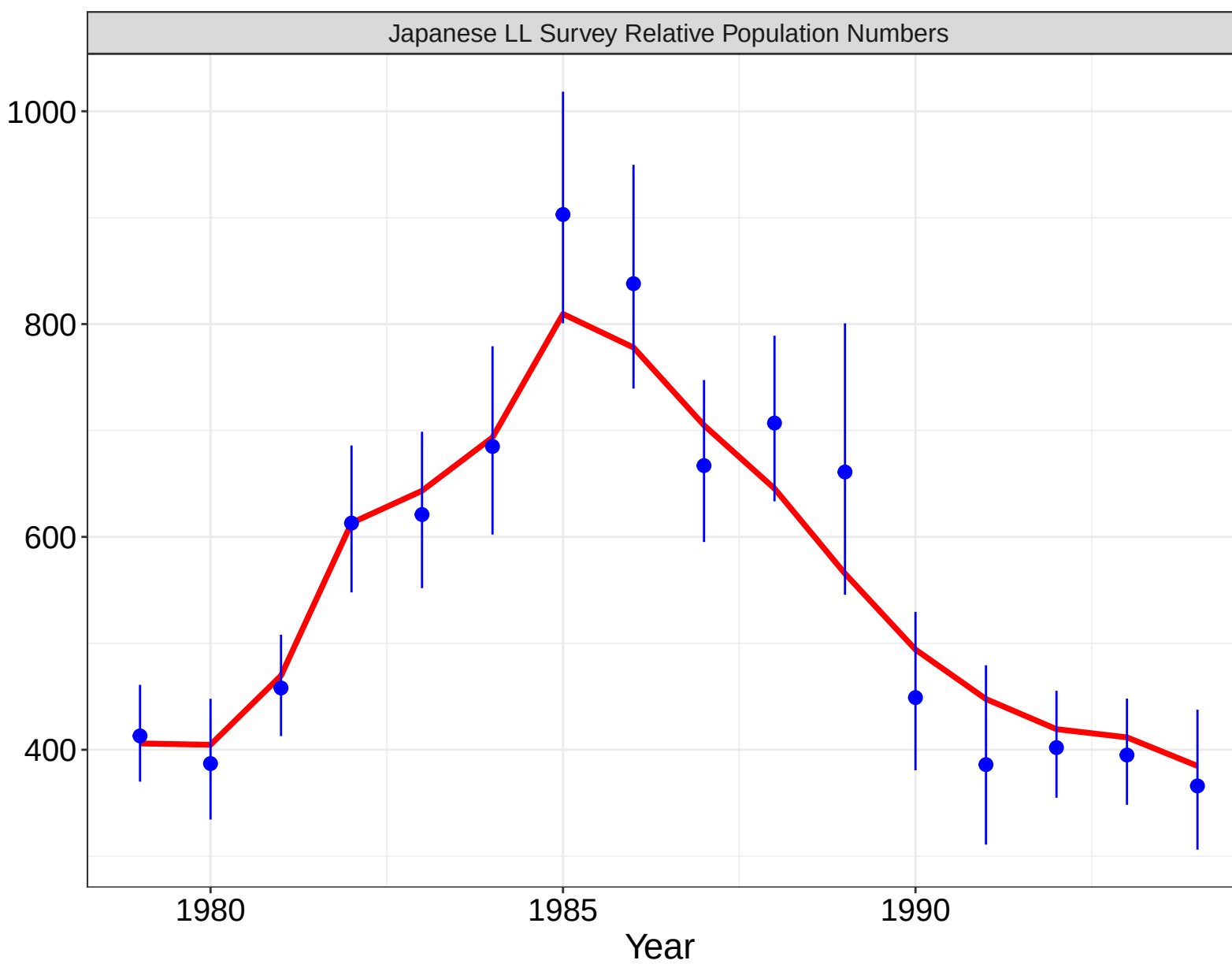
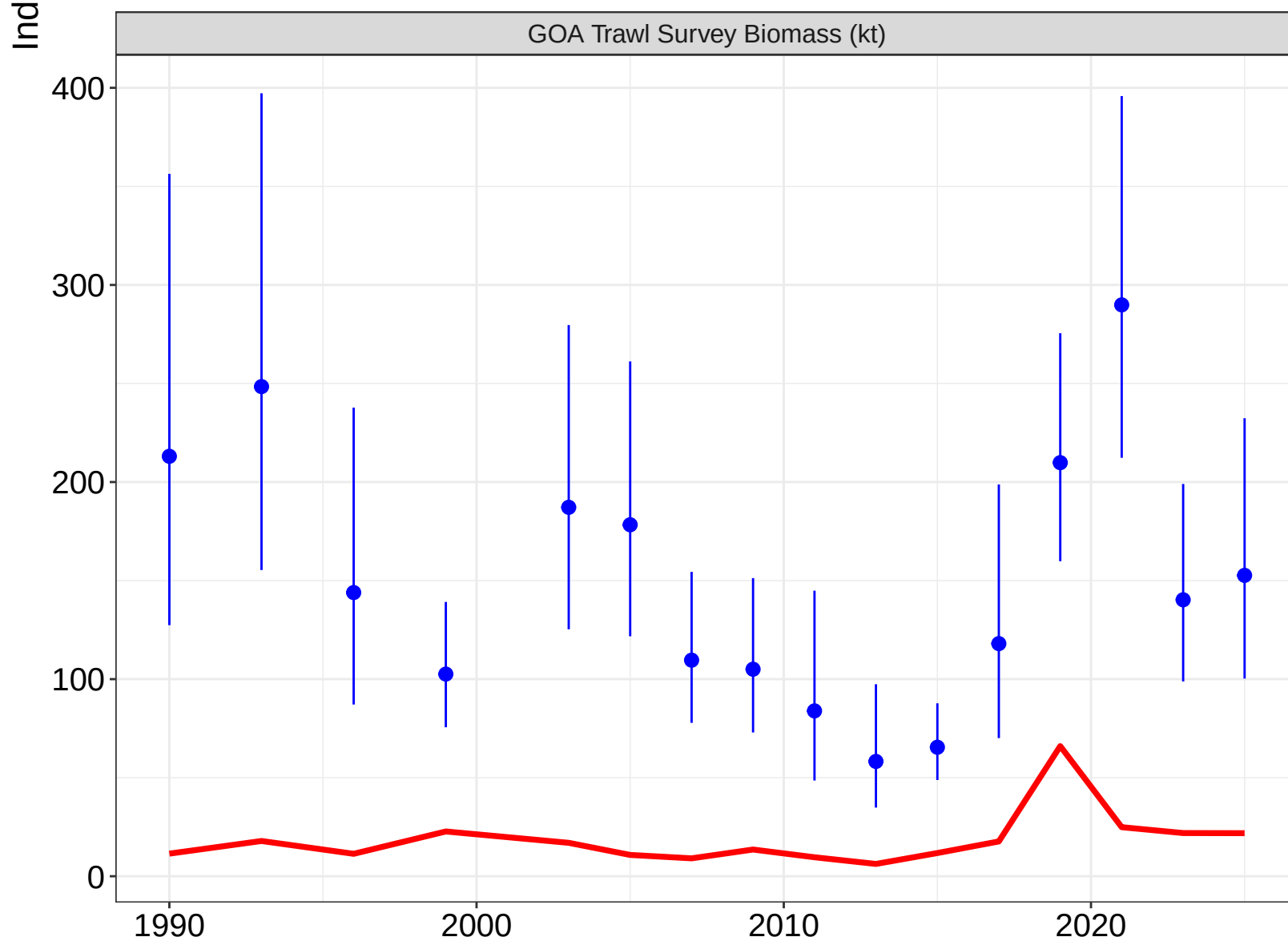
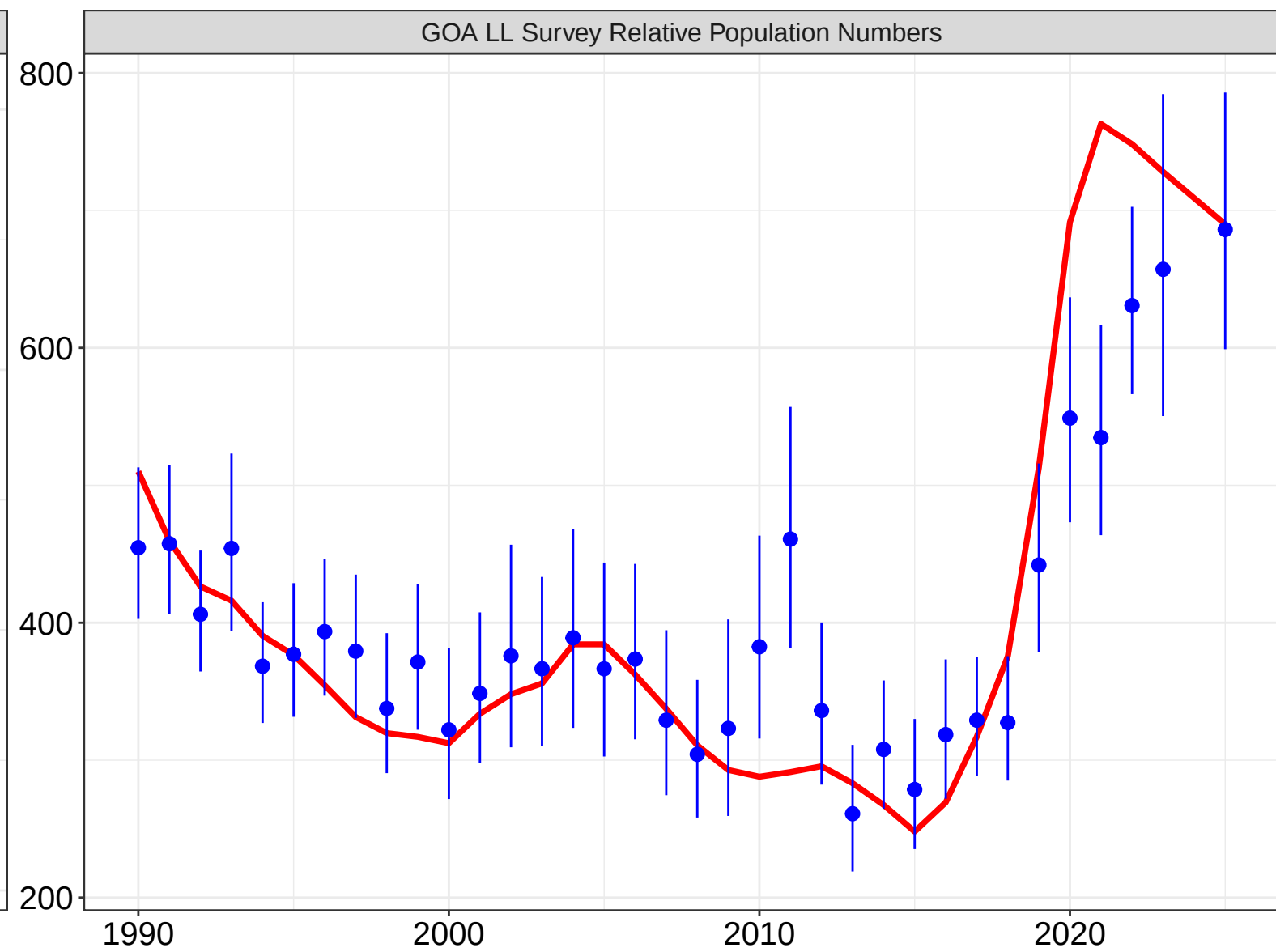
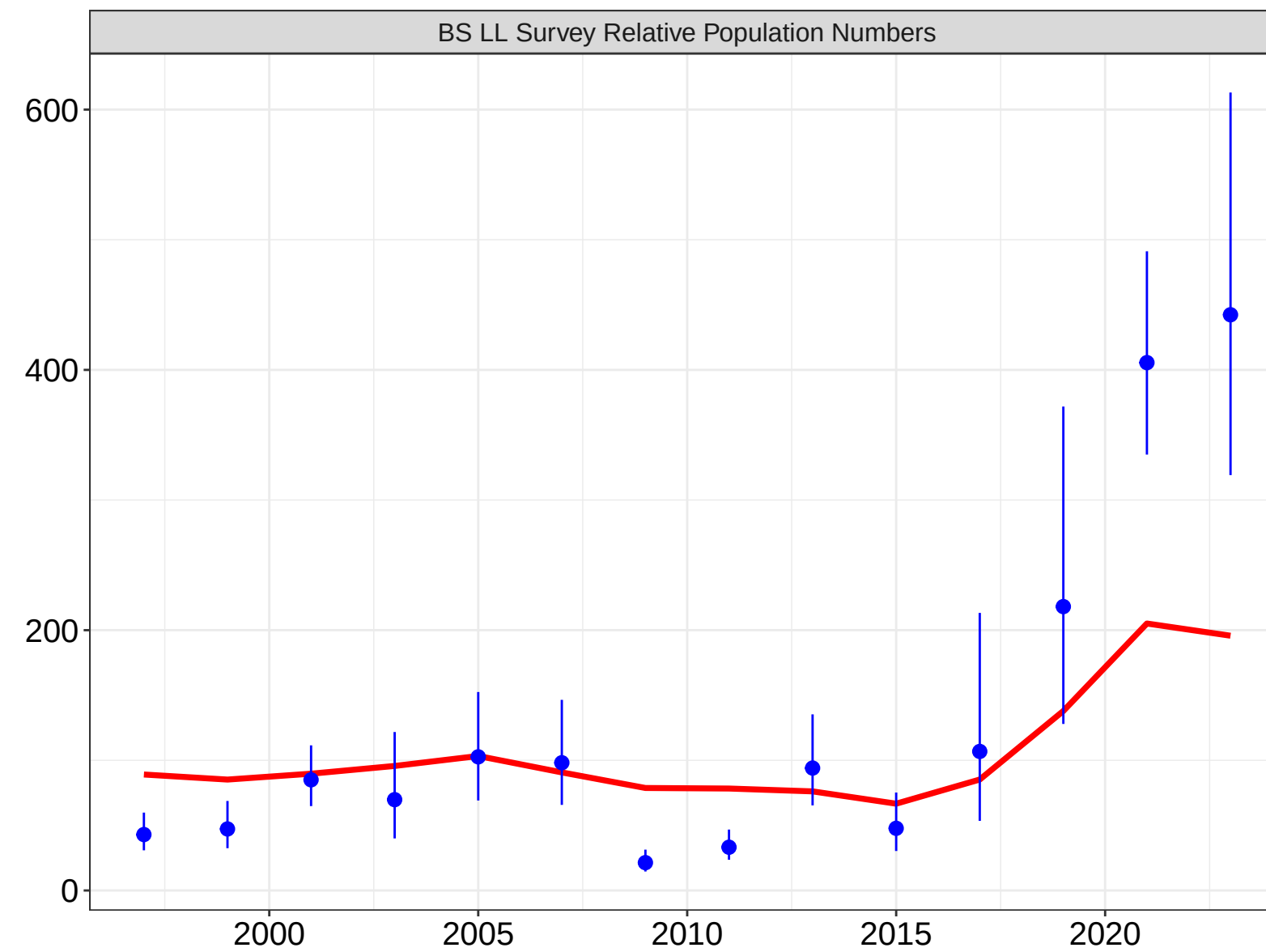
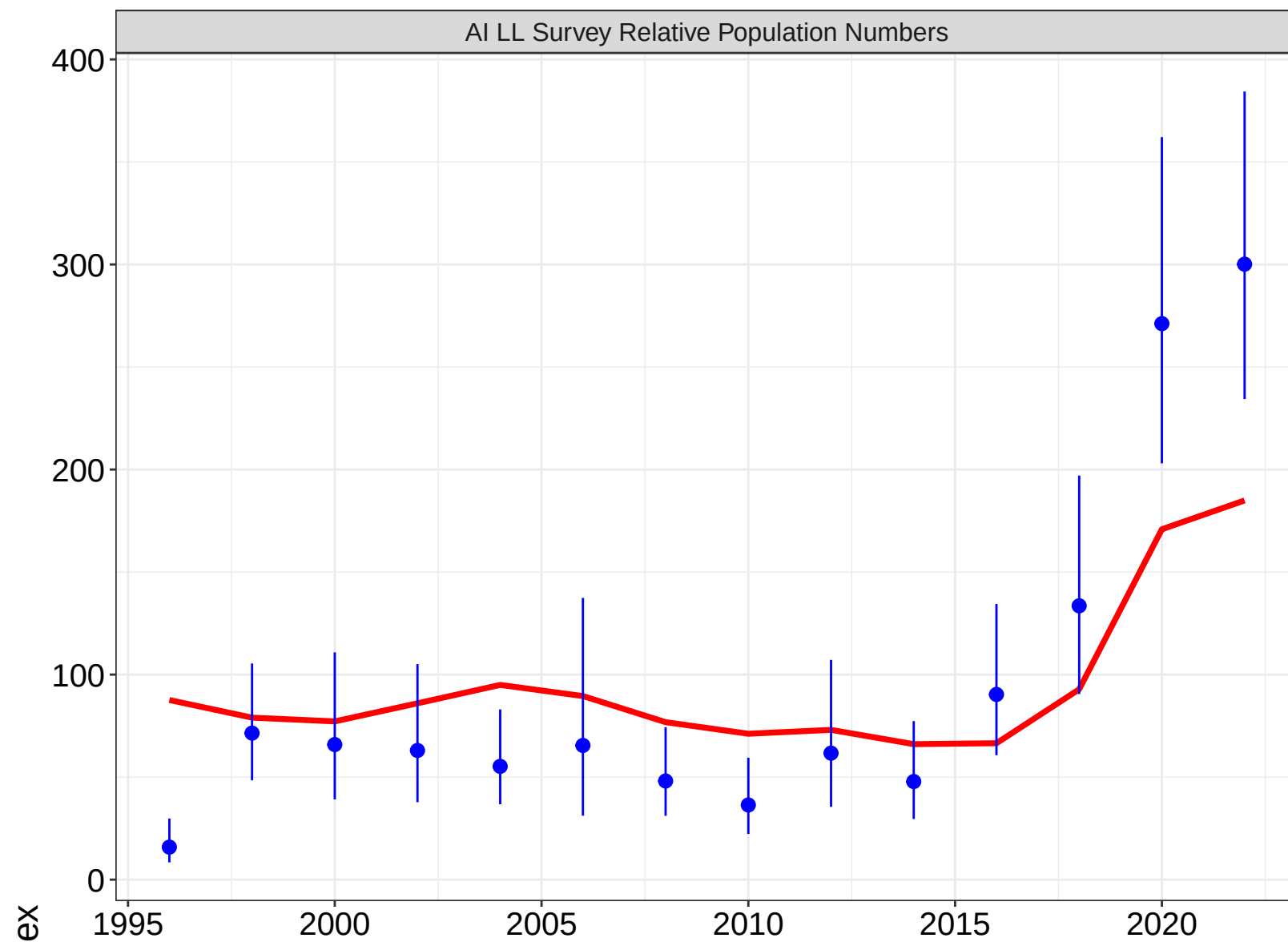
Year



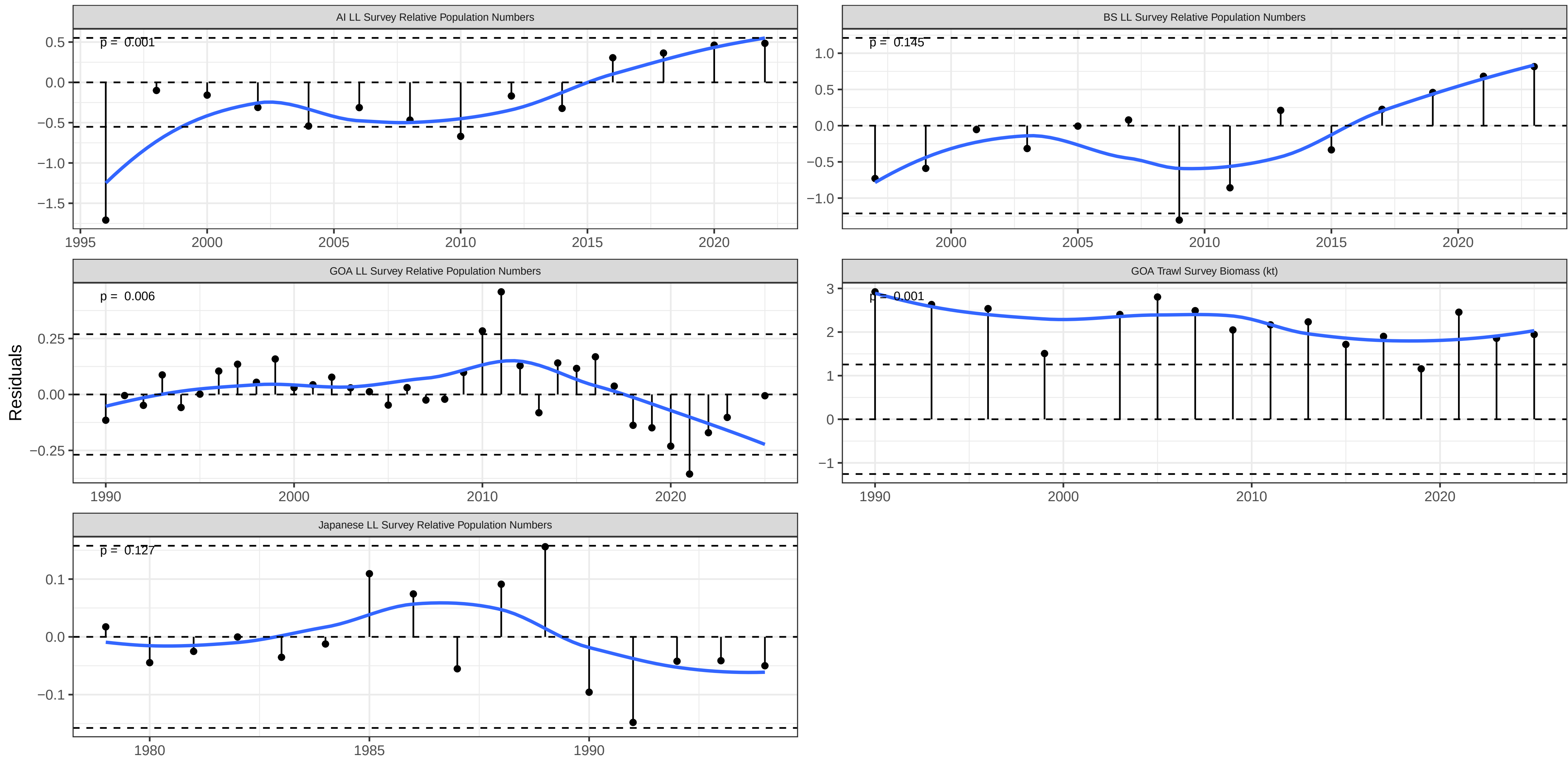




Index Fits

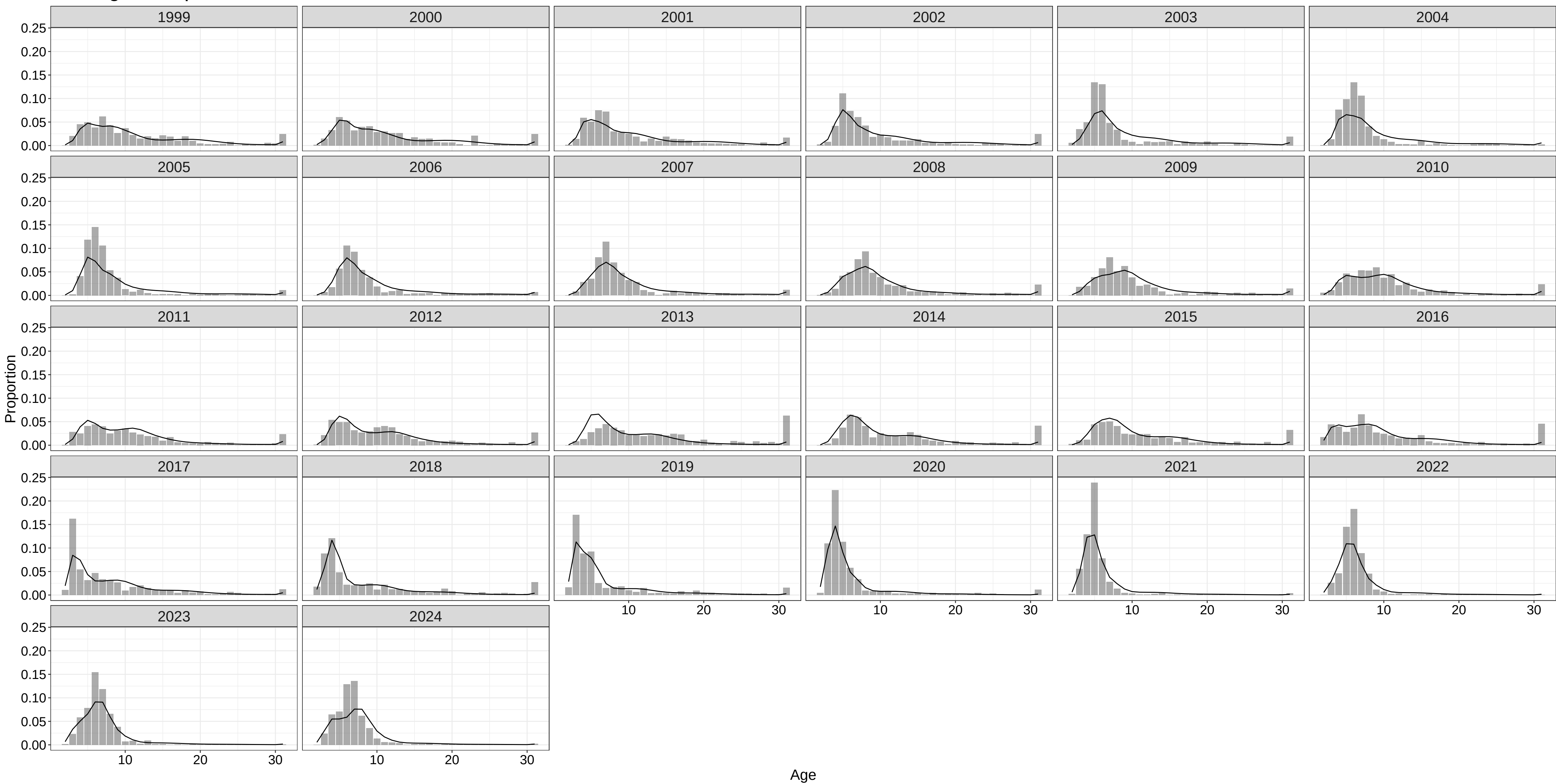


Index Residuals Runs Test

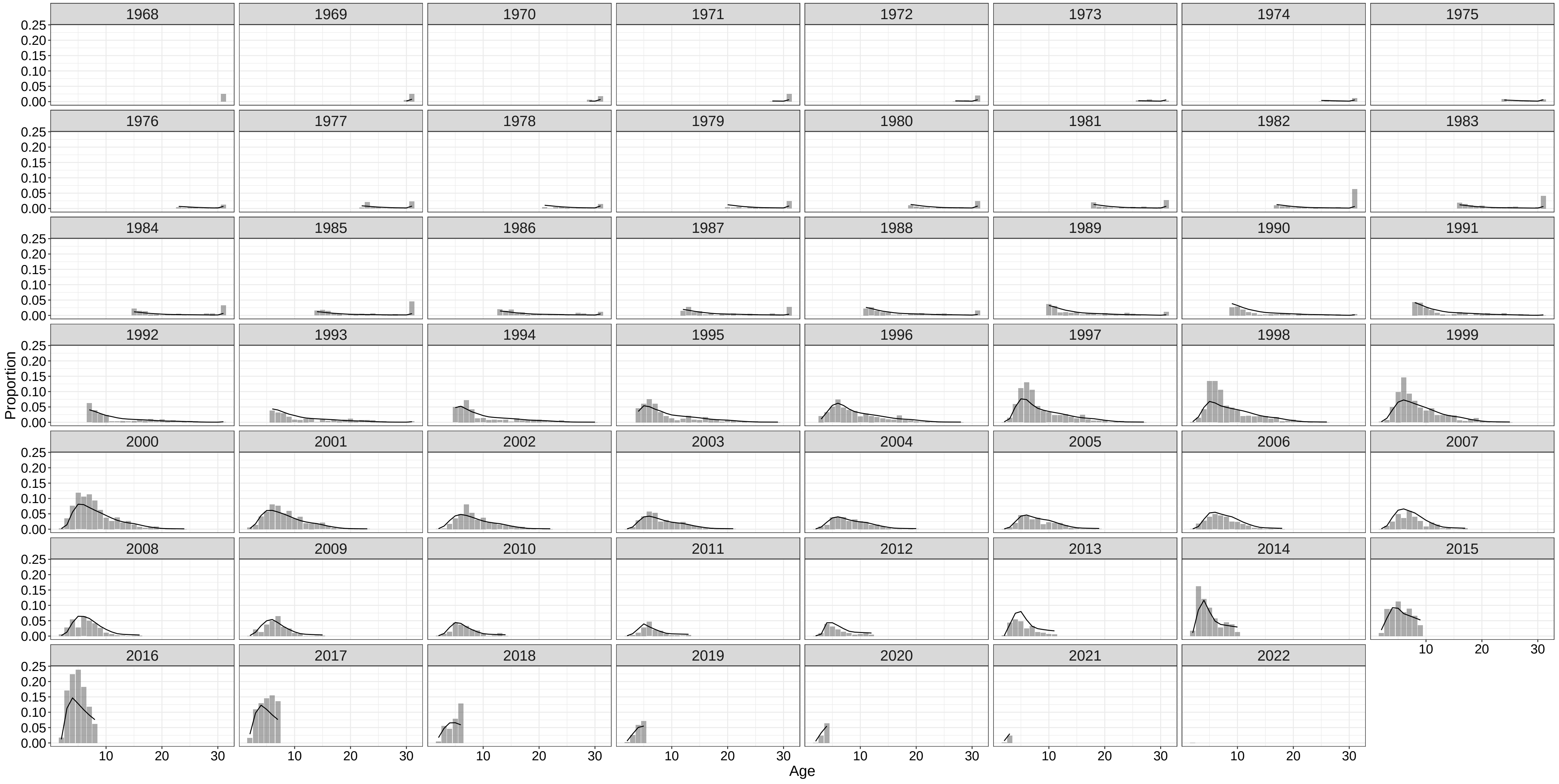


Year

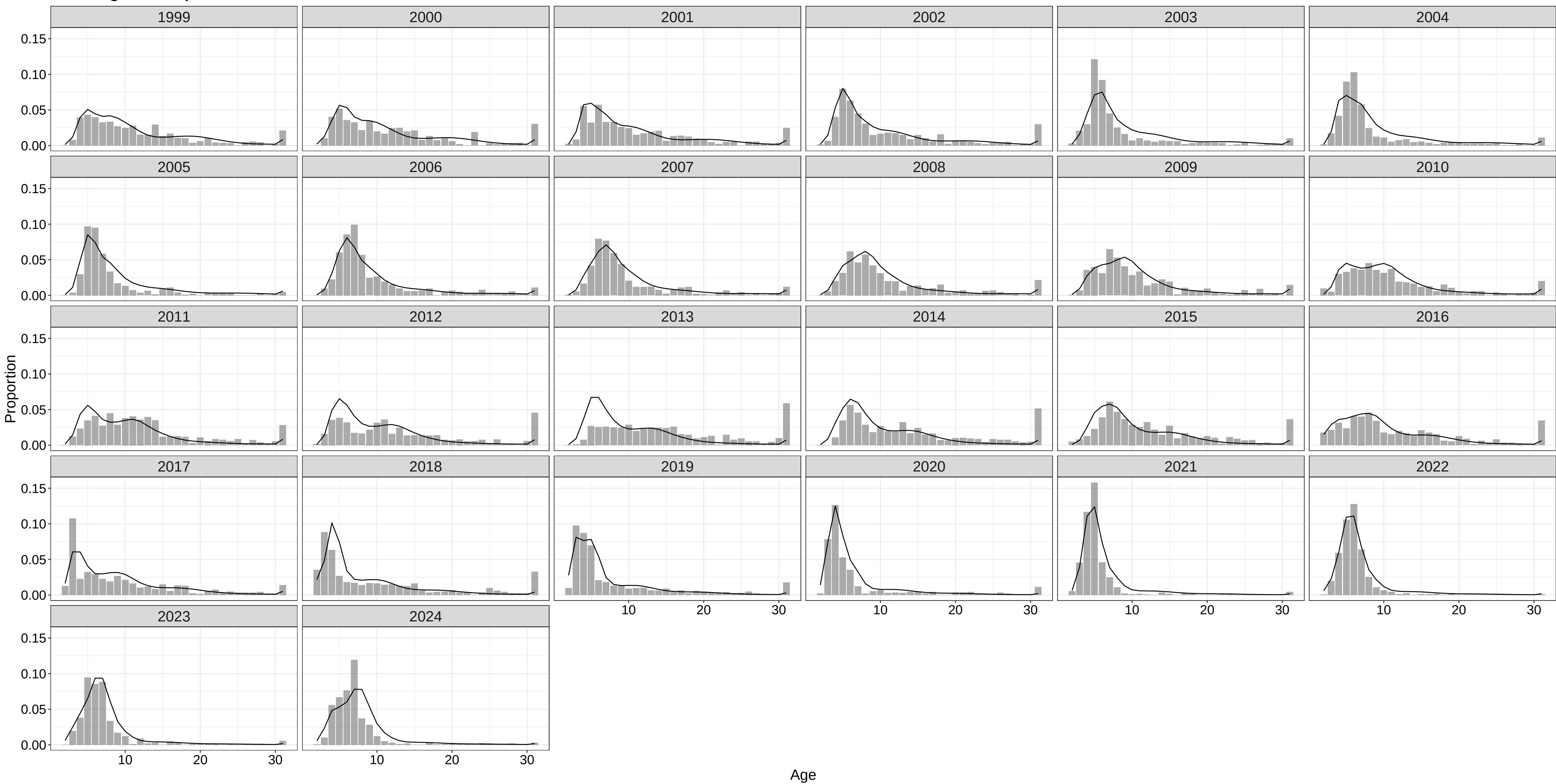
LLF Age Comps Female



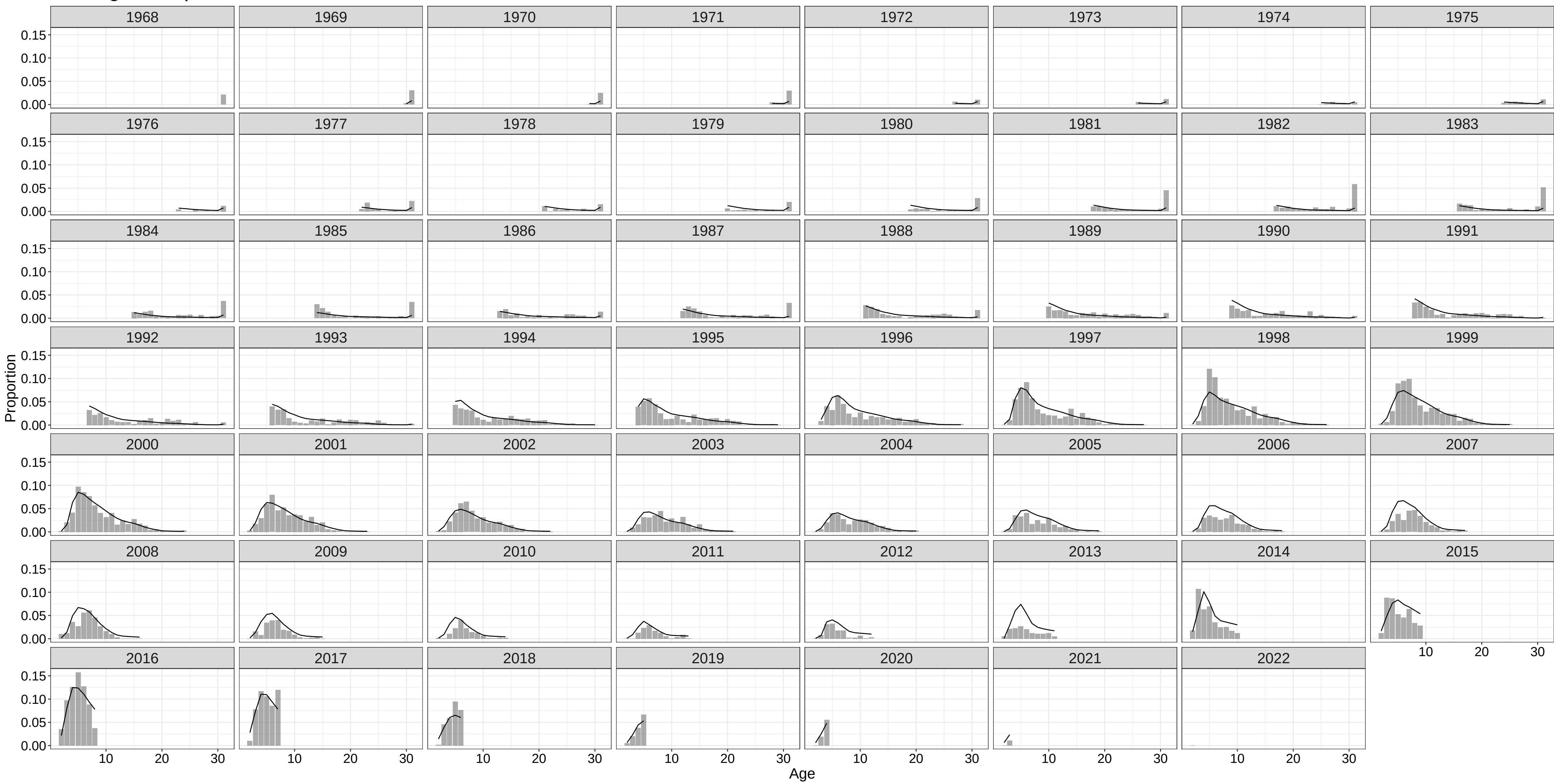
LLF Age Comps Cohort Fits Female



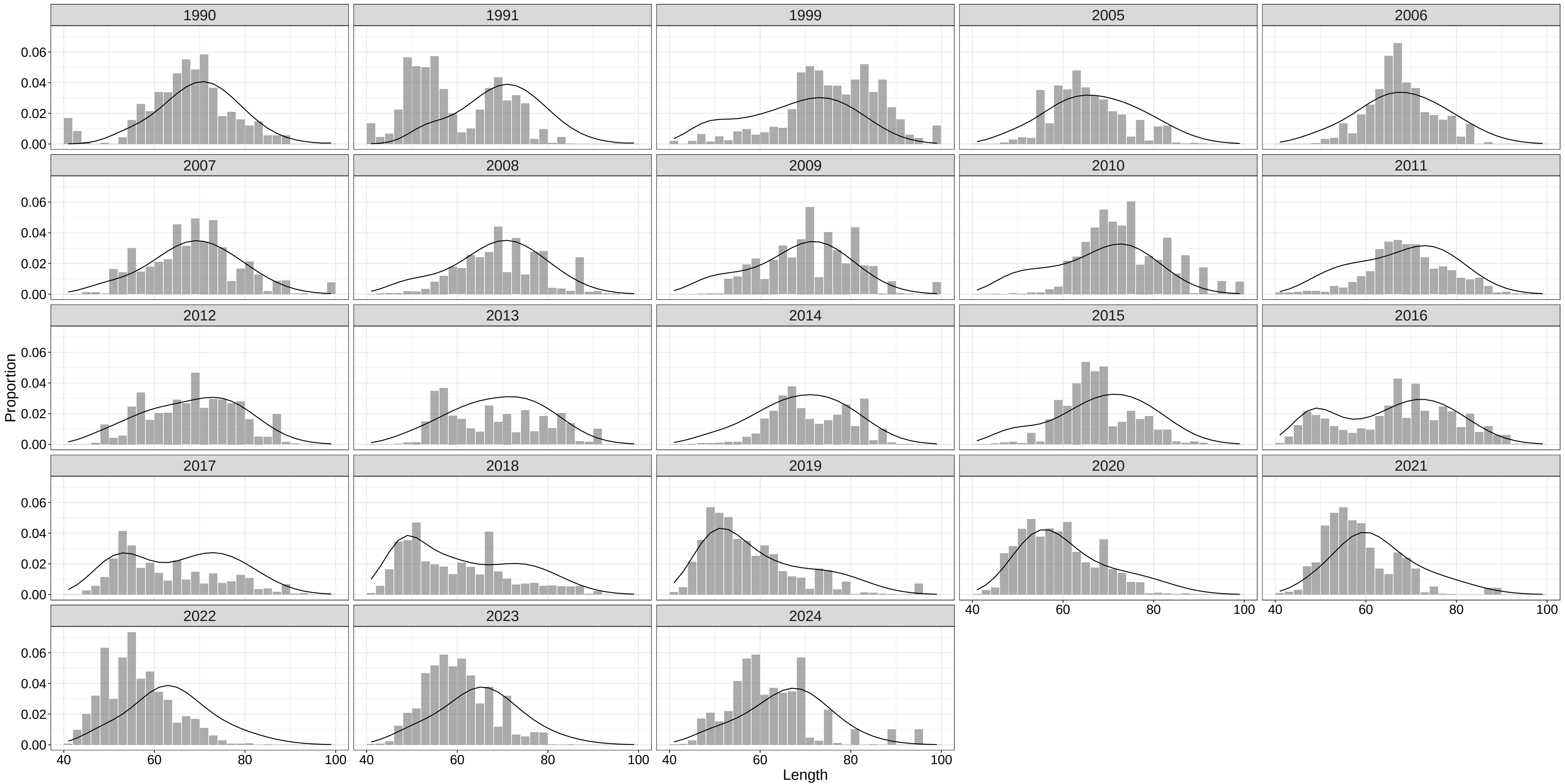
LLF Age Comps Male



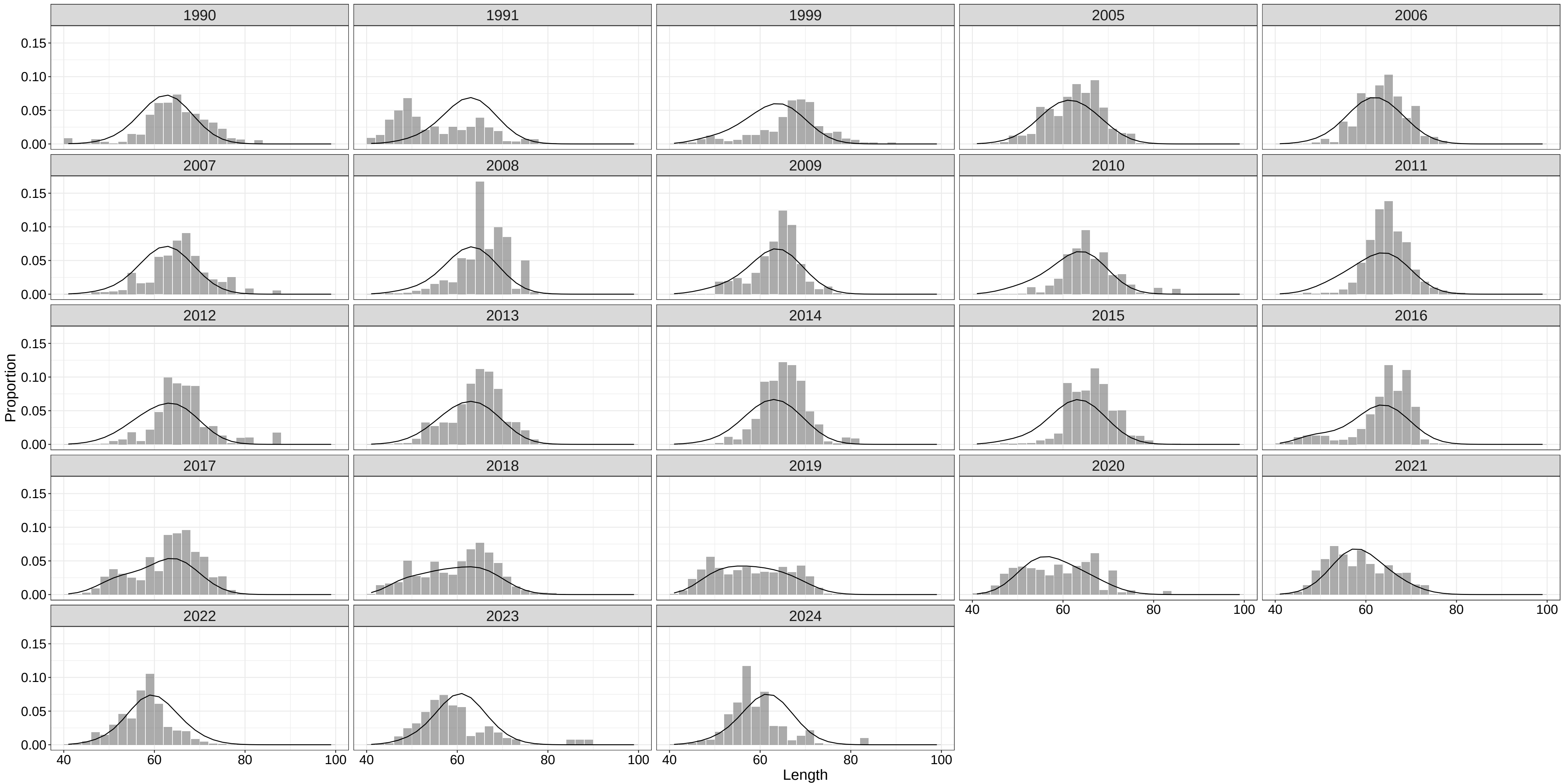
LLF Age Comps Cohort Fits Male



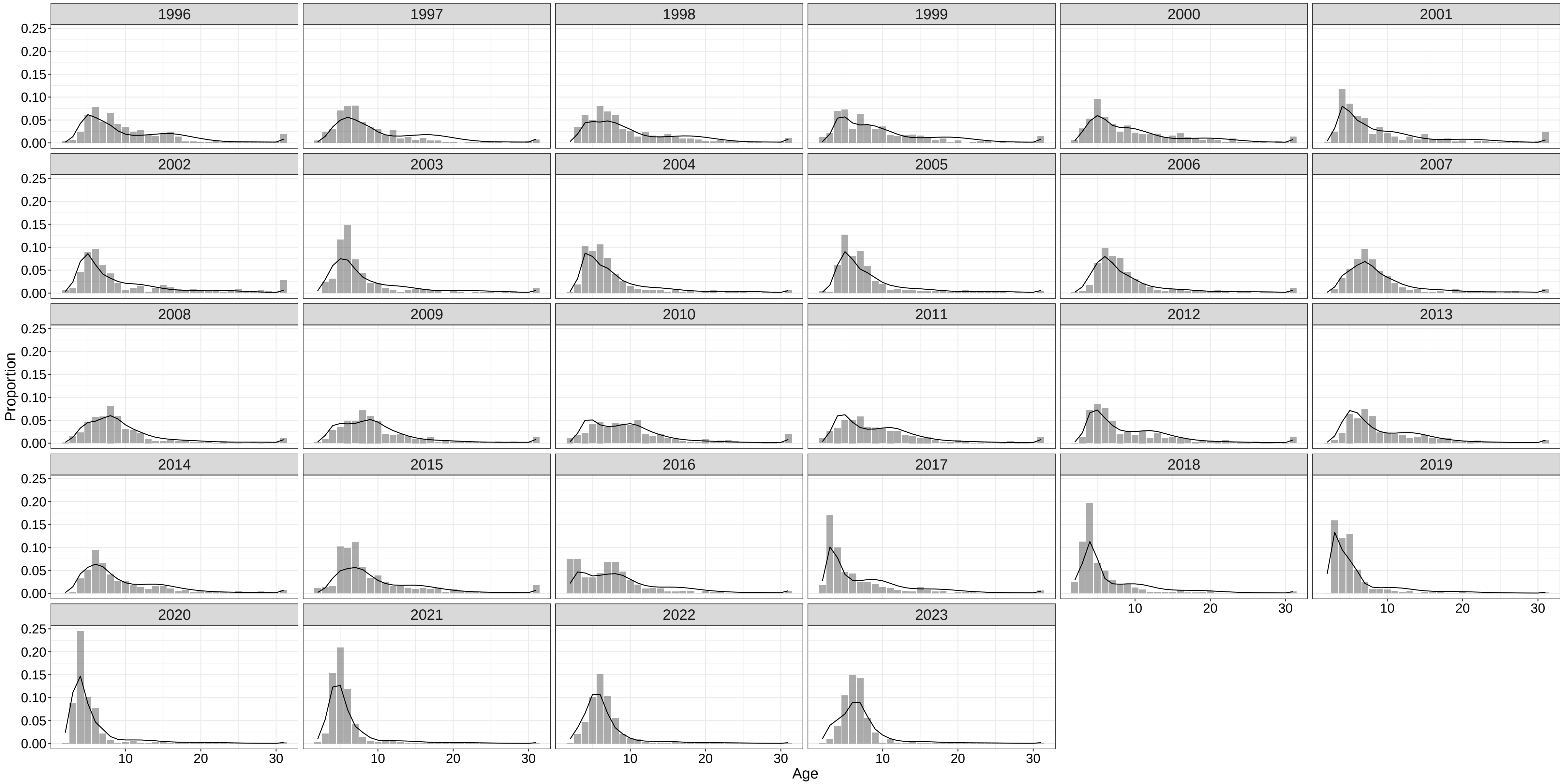
TF Length Comps Female



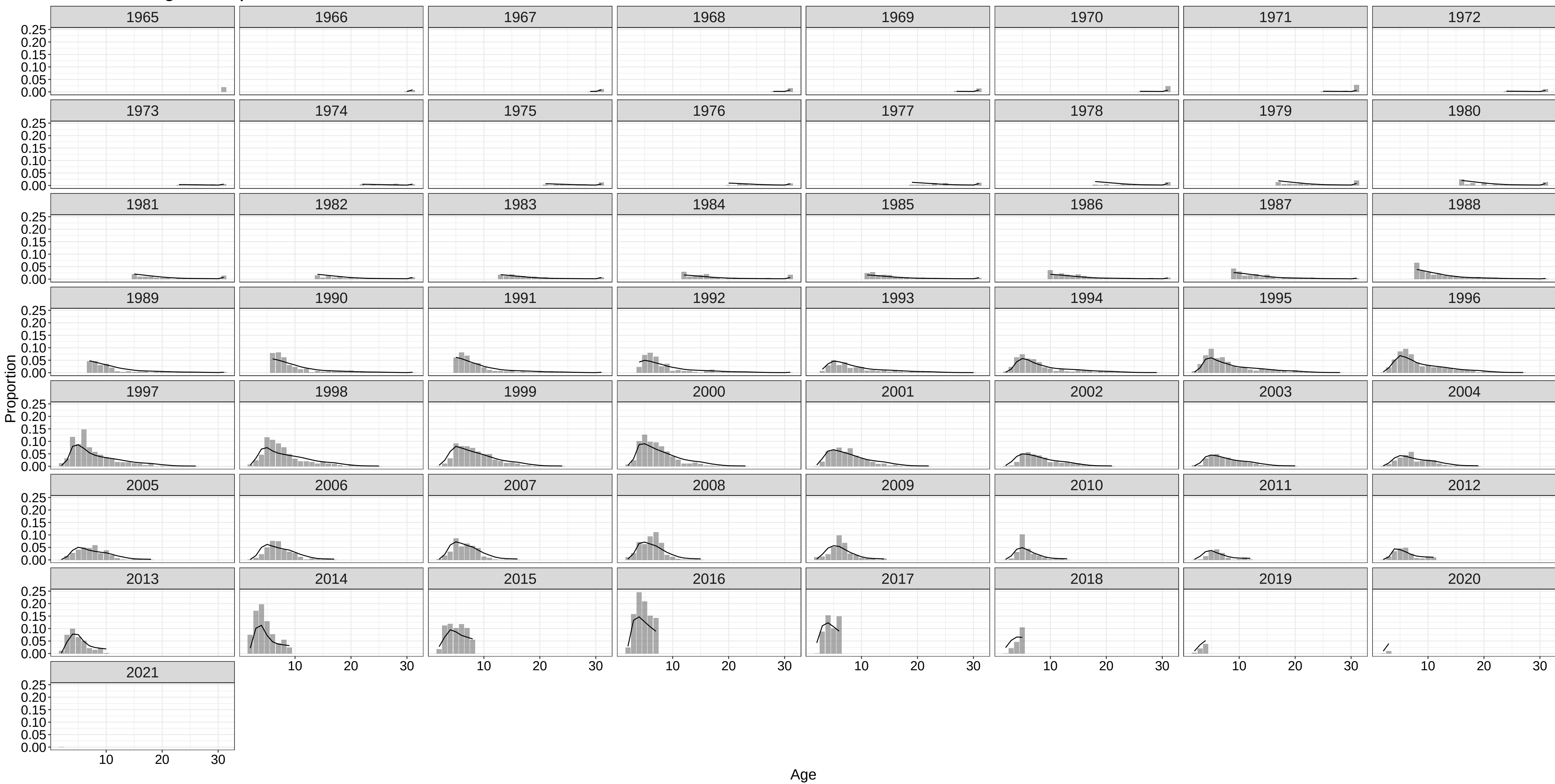
TF Length Comps Male



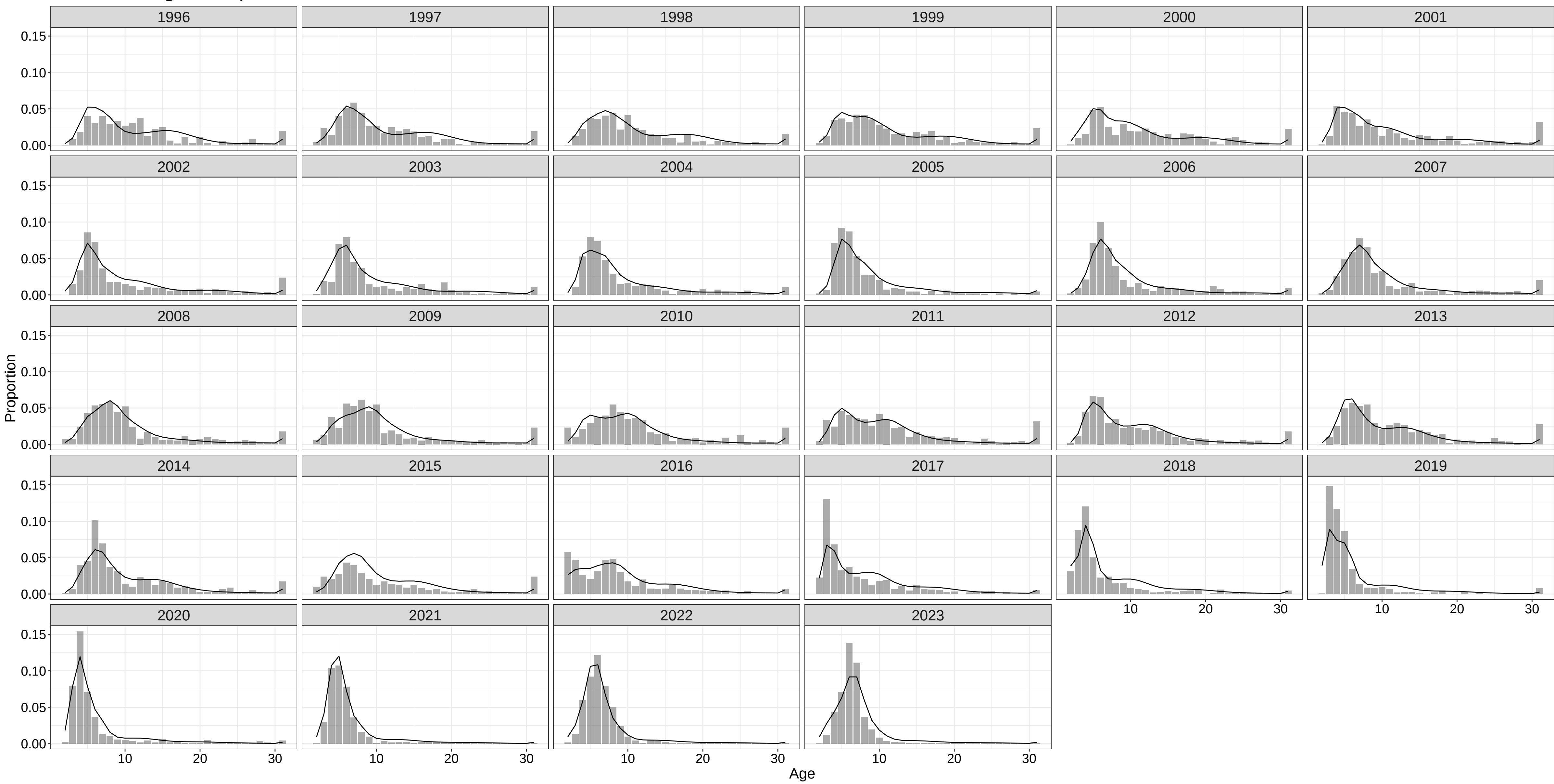
GOA LLS Age Comps Female



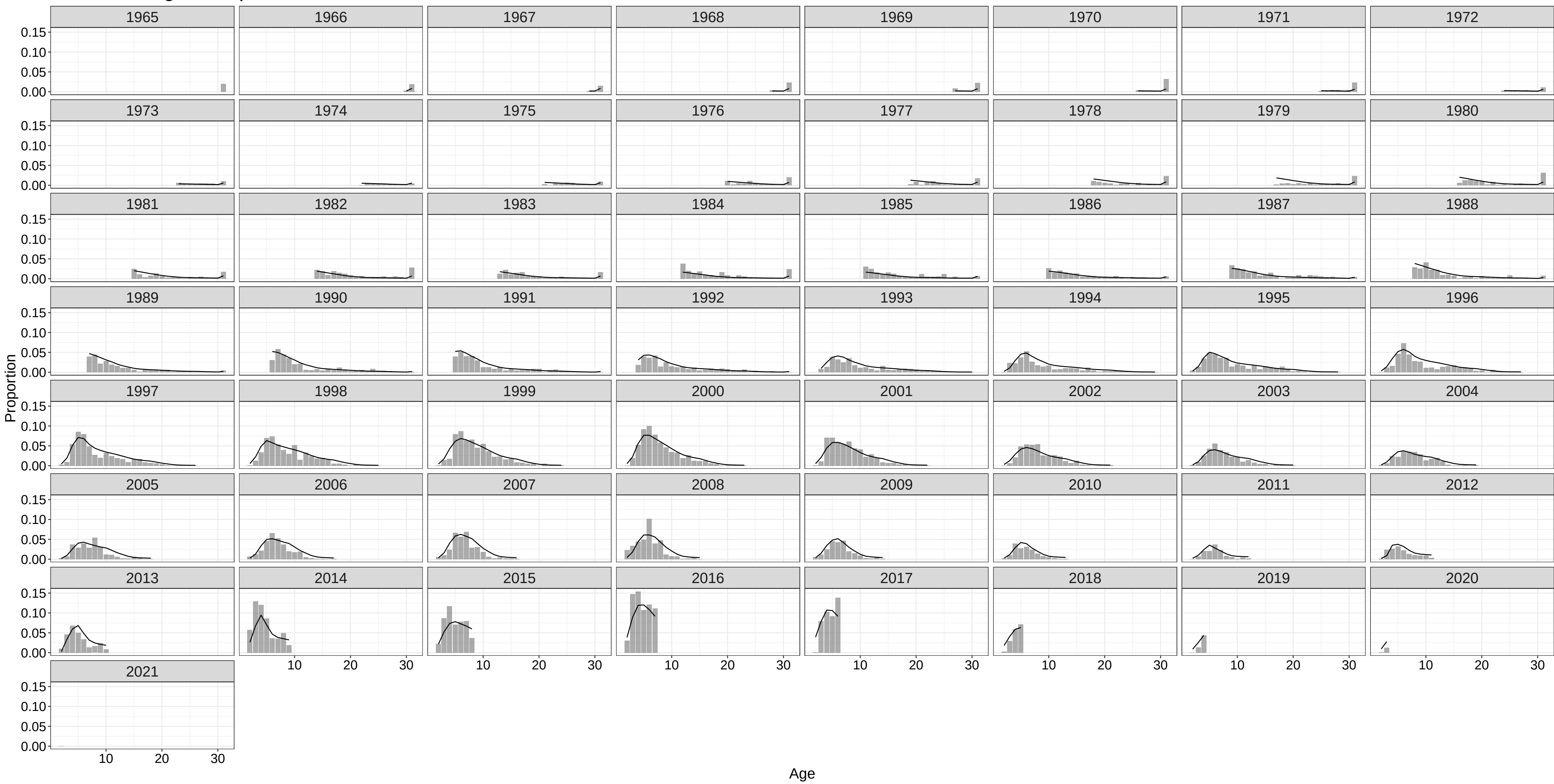
GOA LLS Age Comps Cohort Fit Female



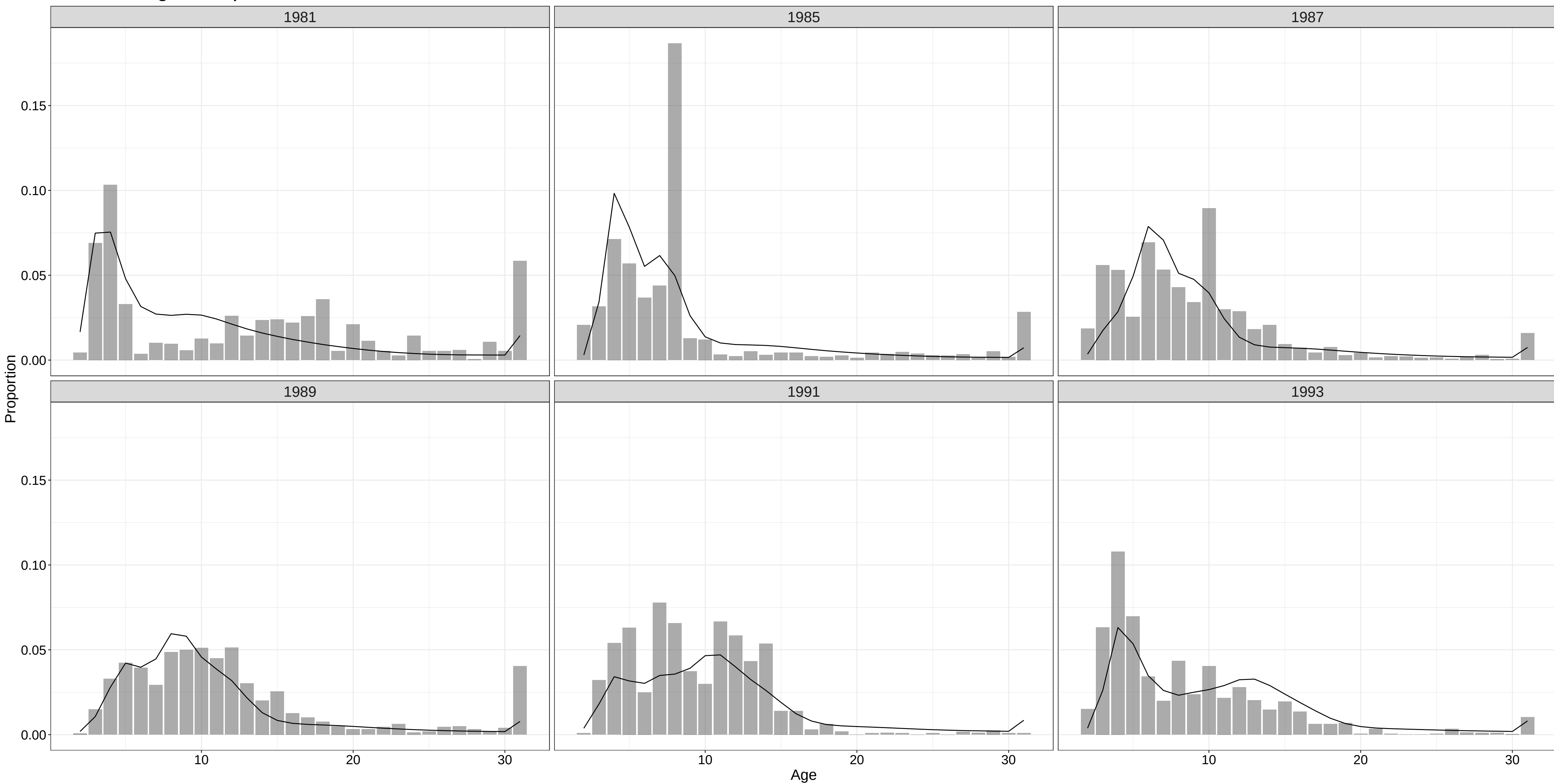
GOA LLS Age Comps Male



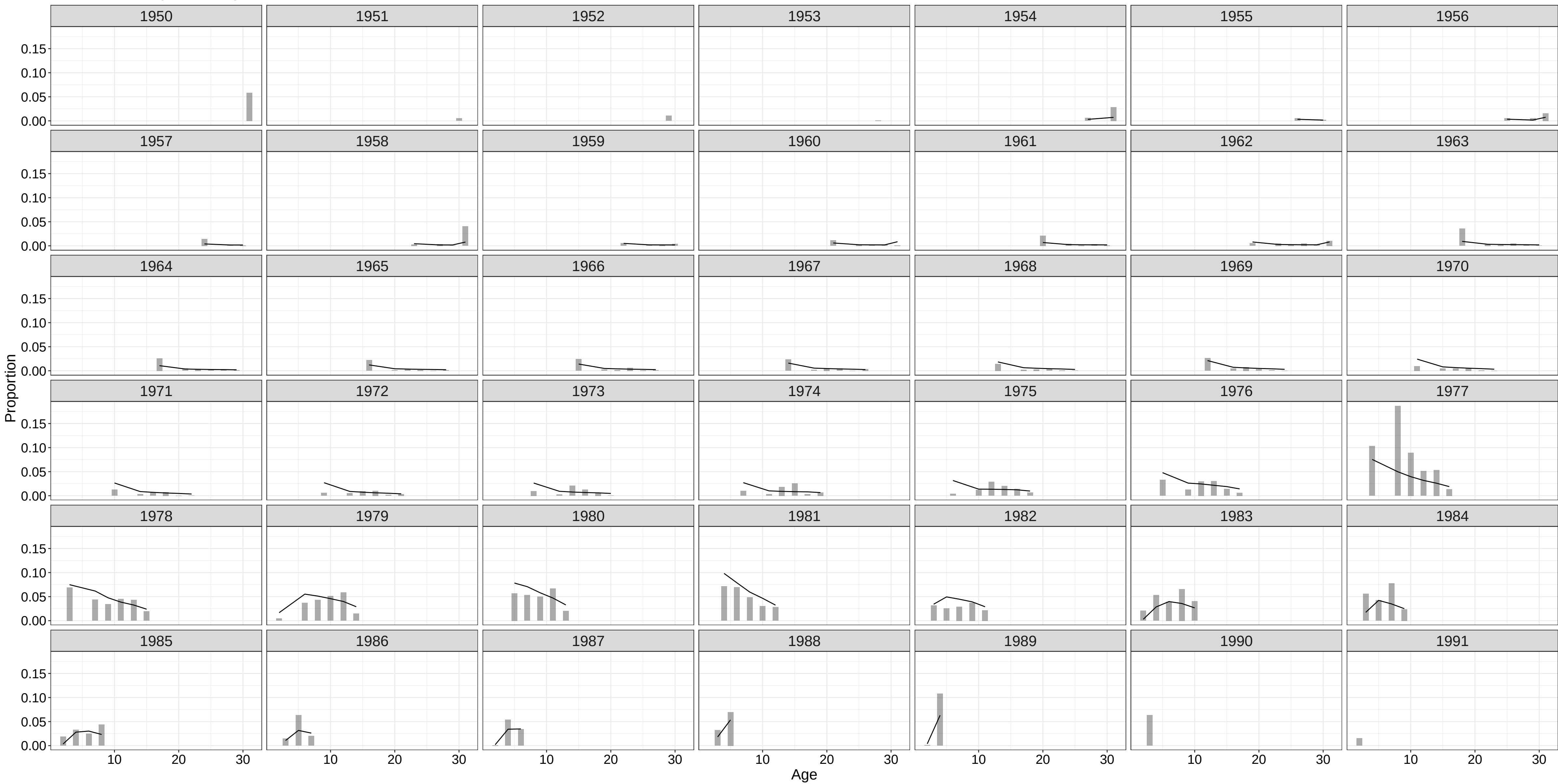
GOA LLS Age Comps Cohort Fit Male



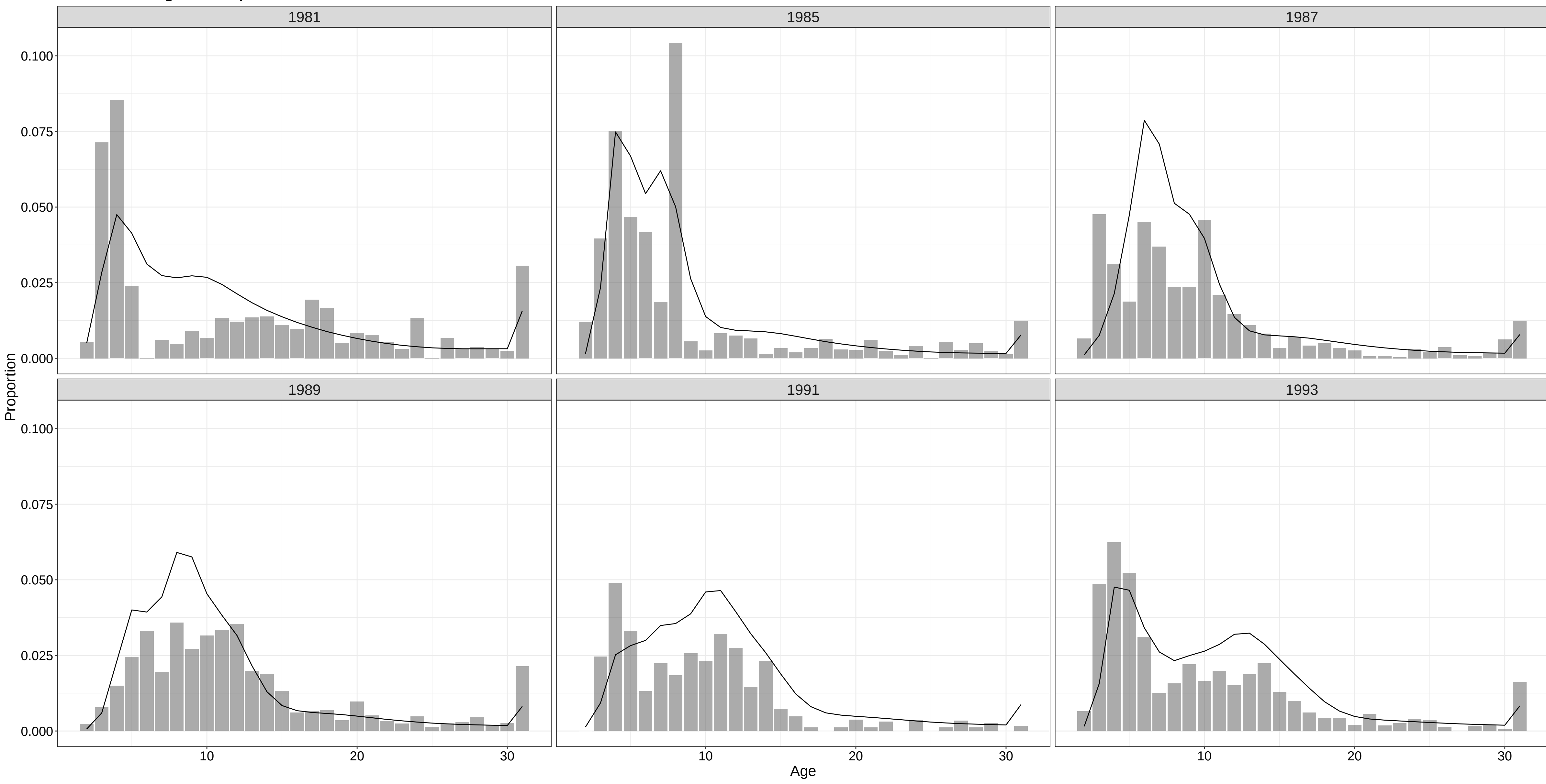
JPN LLS Age Comps Female



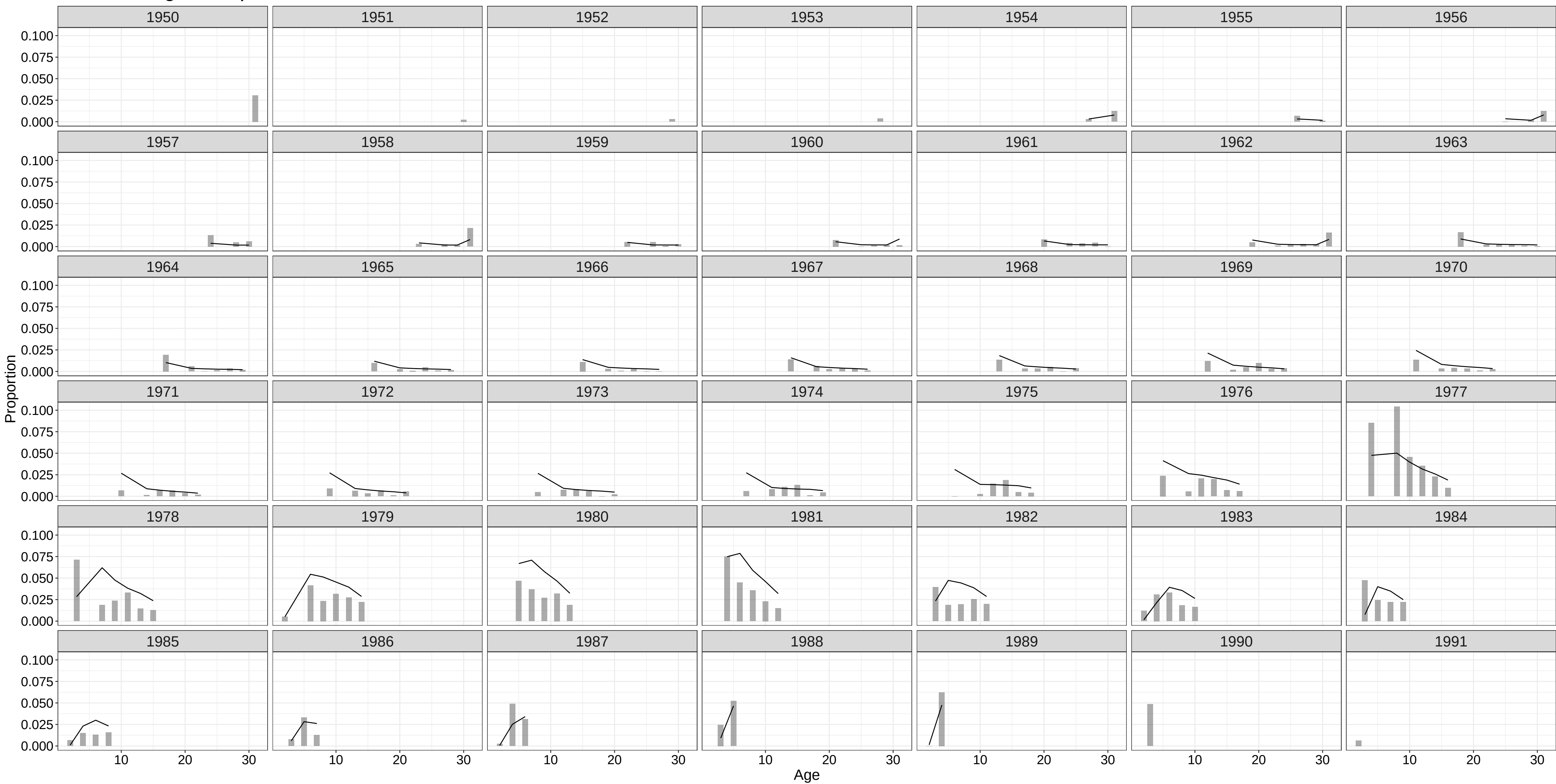
JPN LLS Age Comps Cohort Fit Female



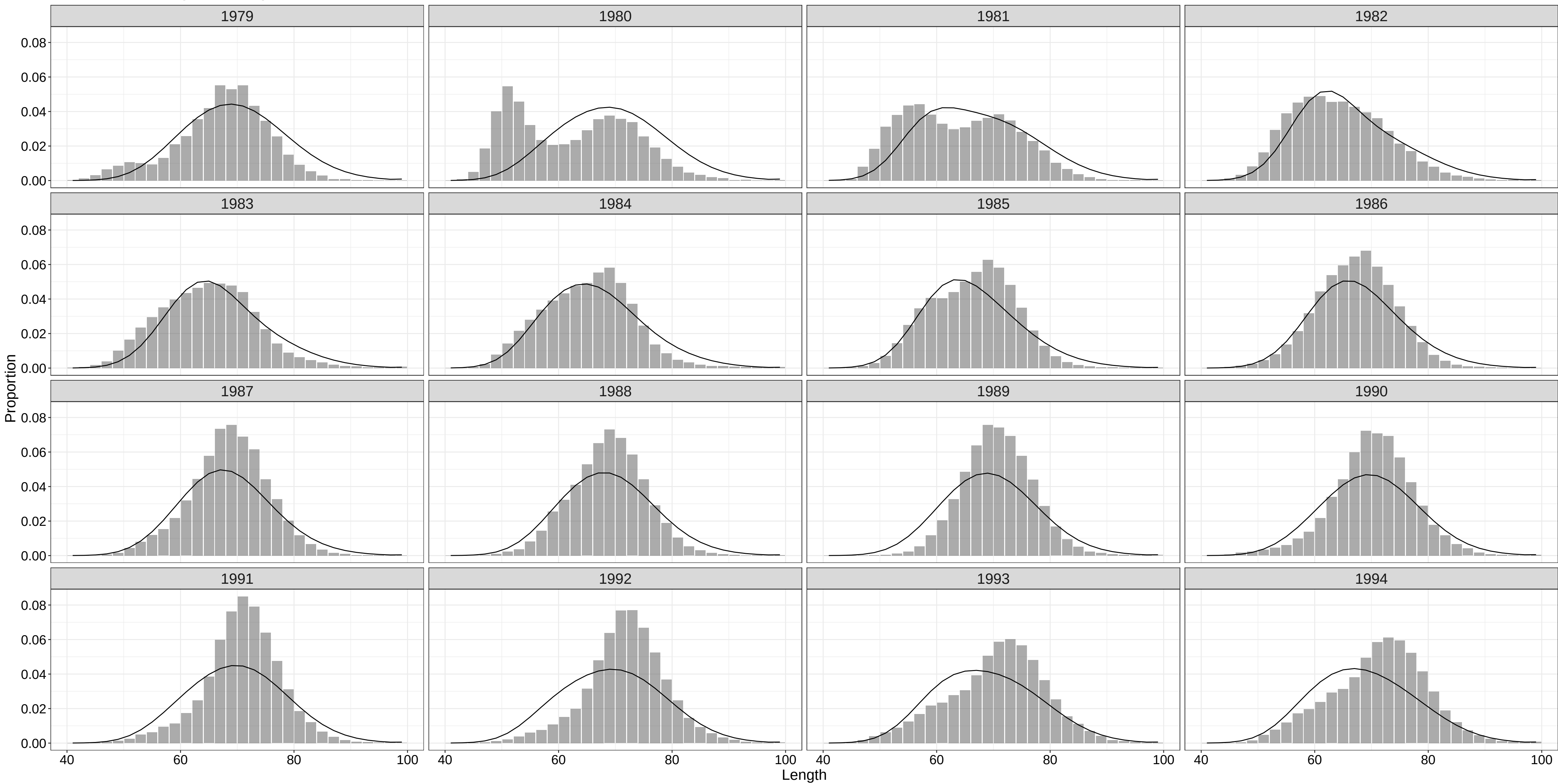
JPN LLS Age Comps Male



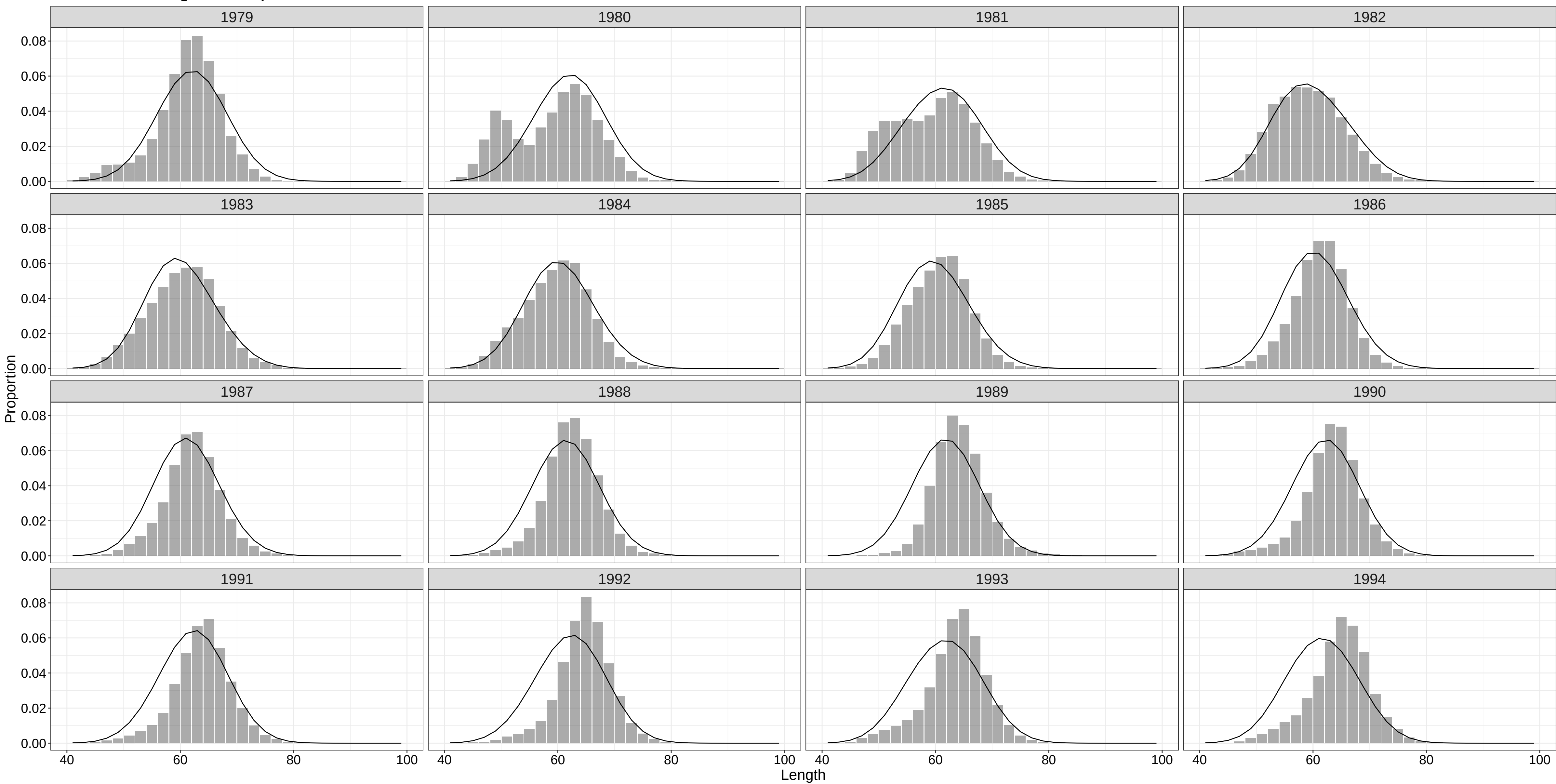
JPN LLS Age Comps Cohort Fit Male



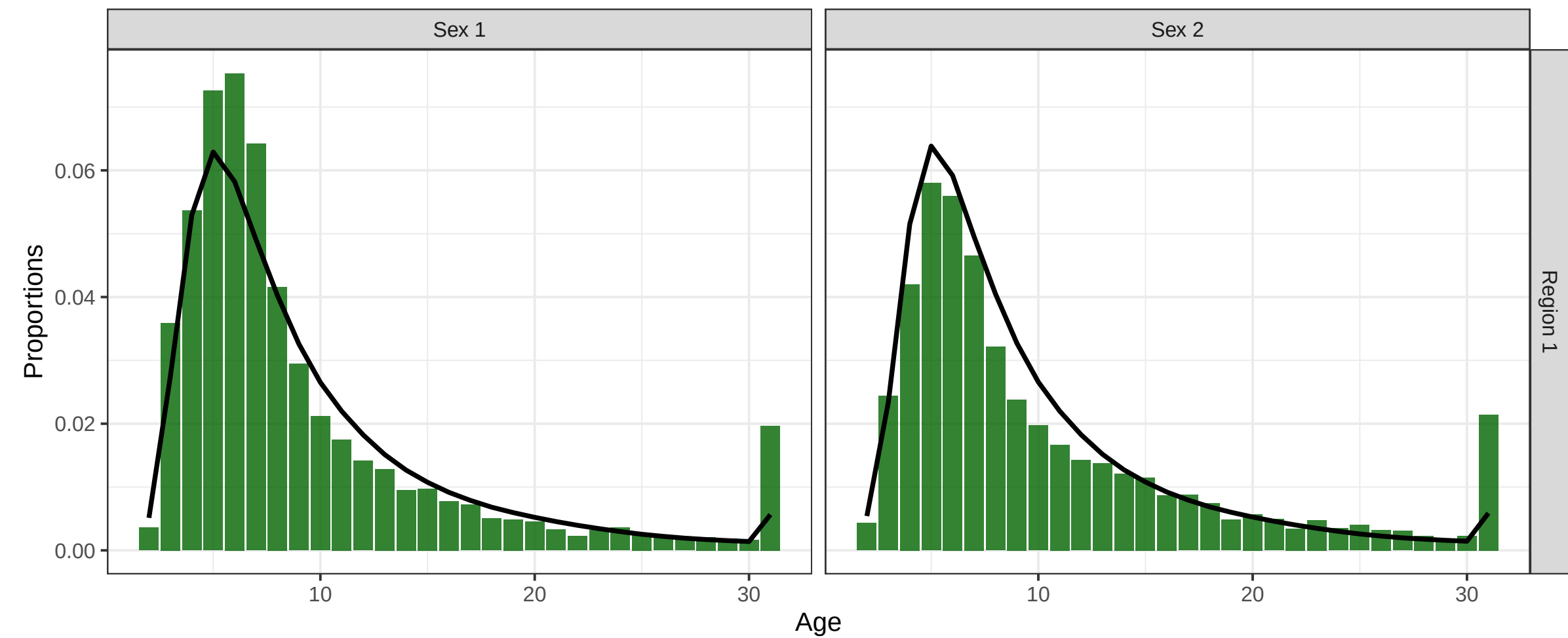
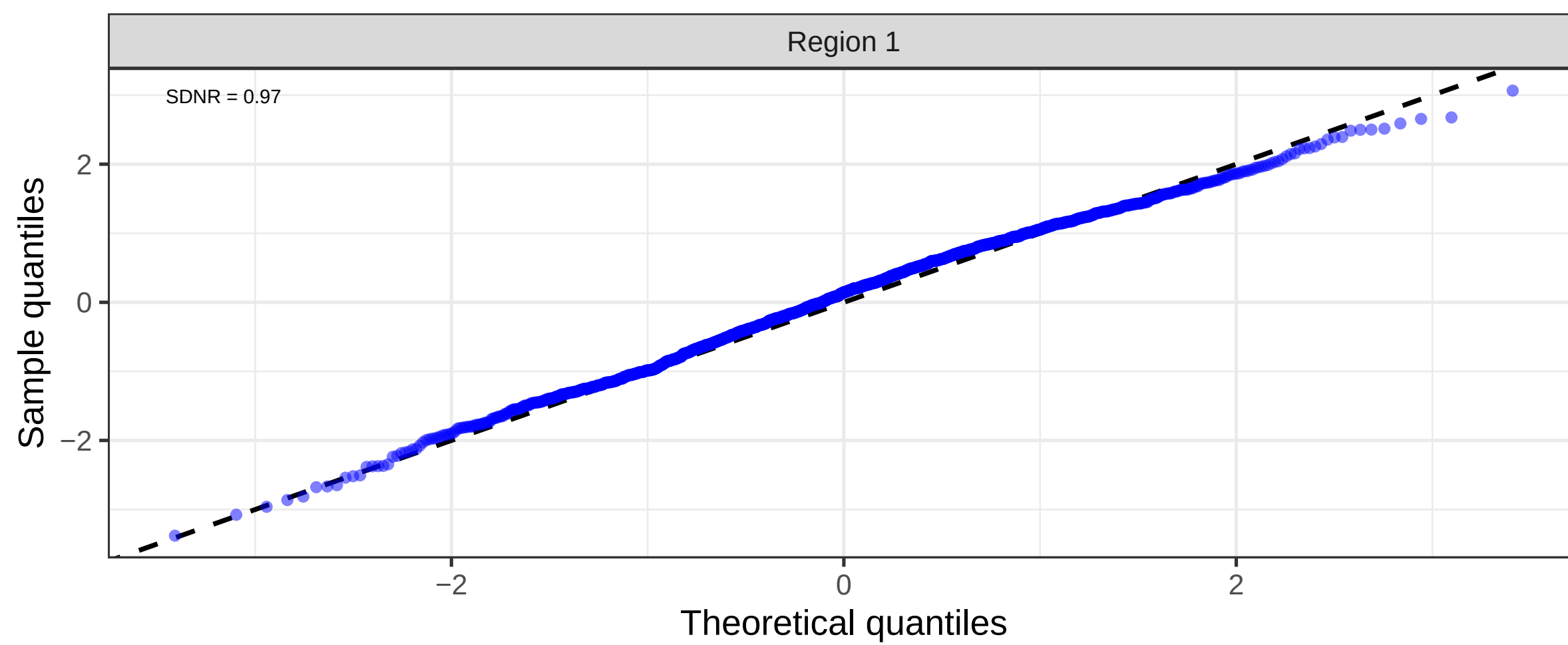
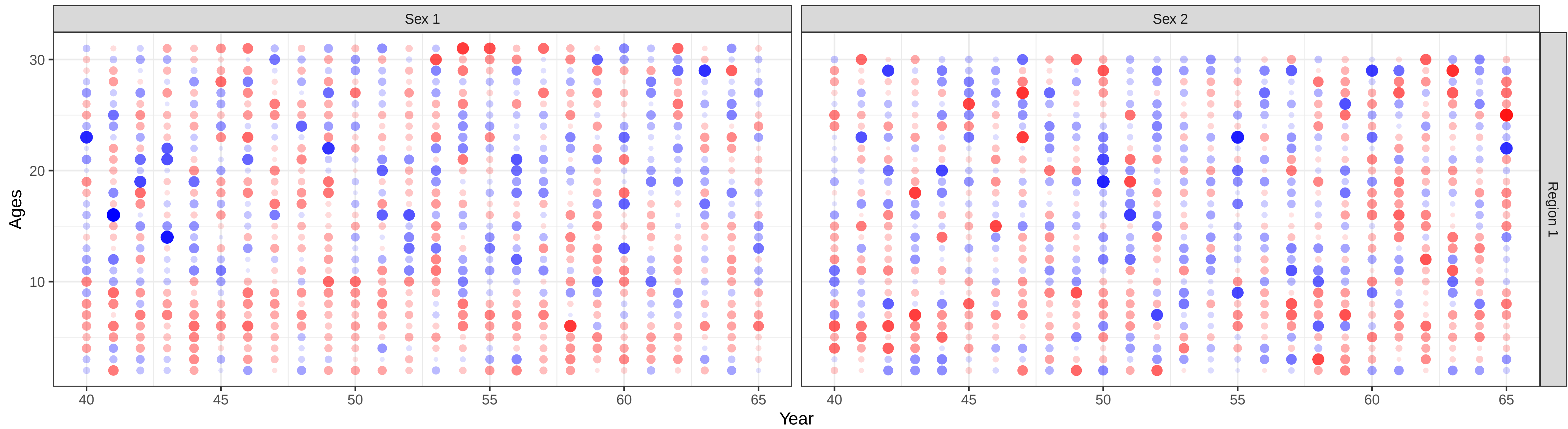
JPN LLS Length Comps Female

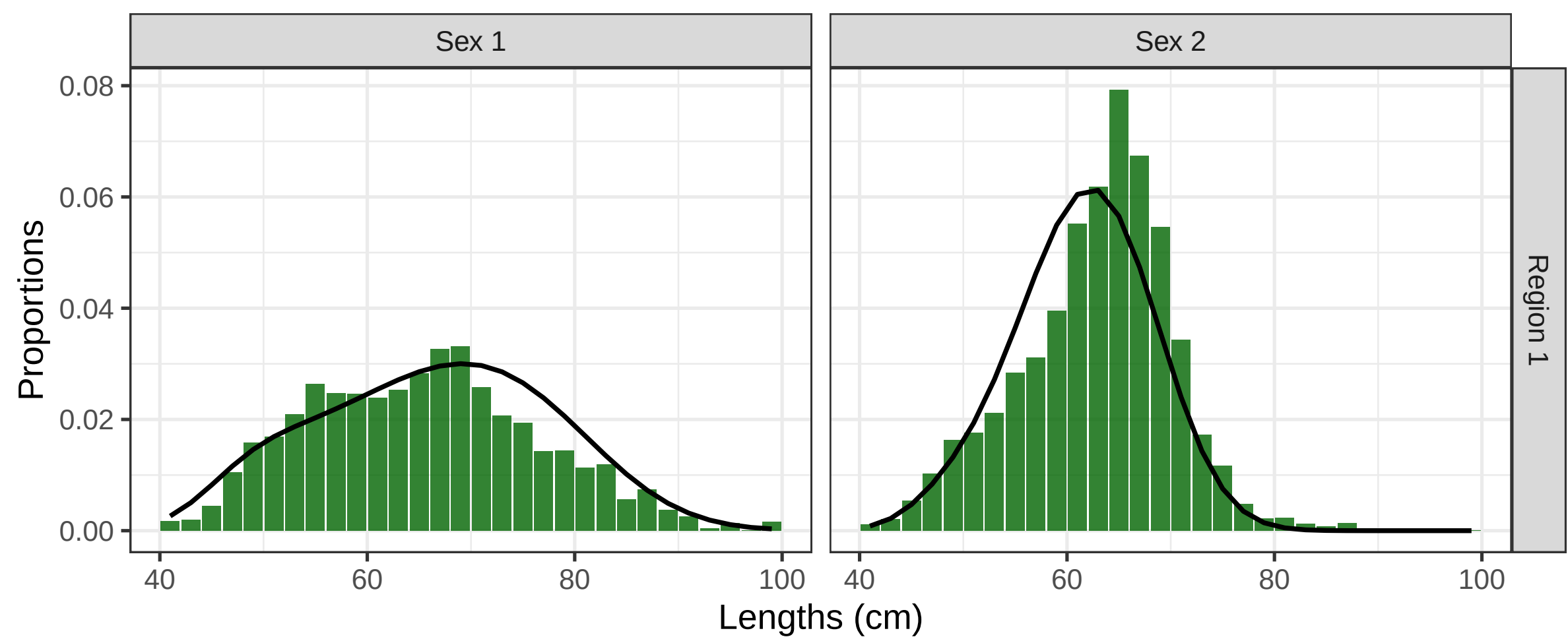
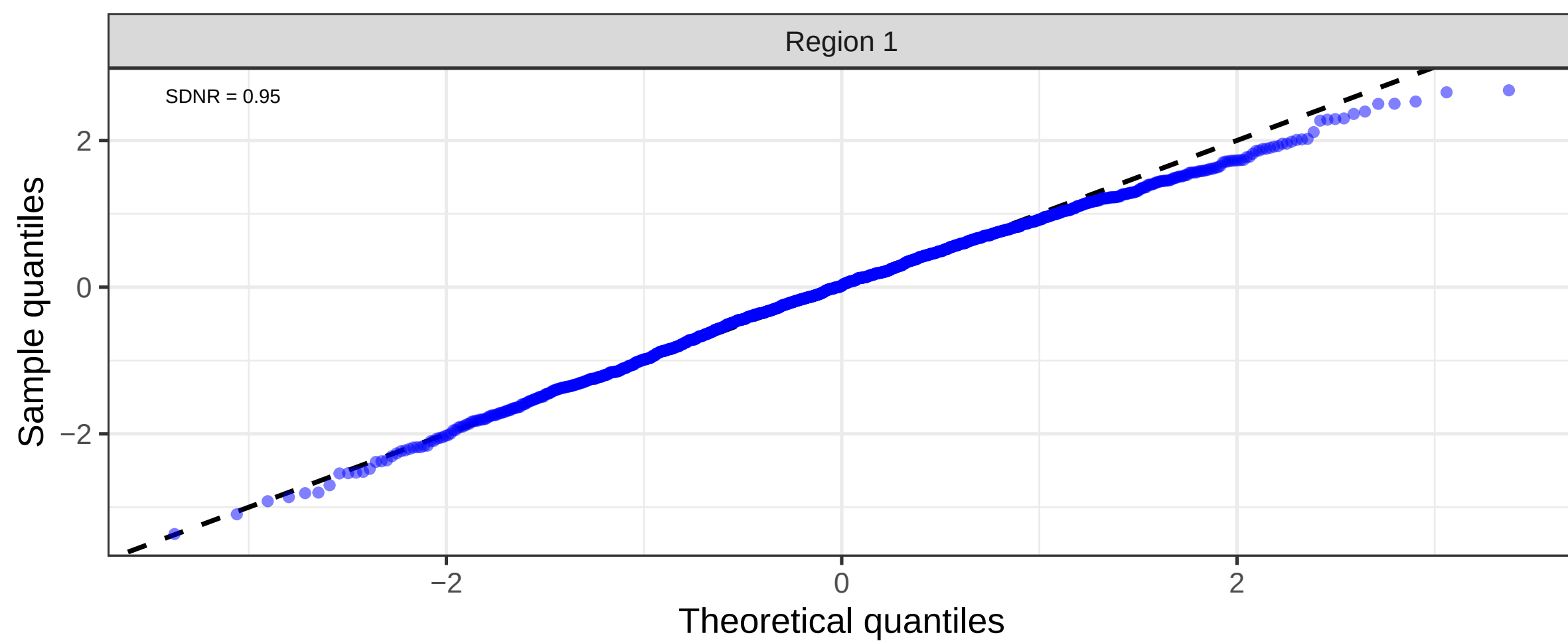
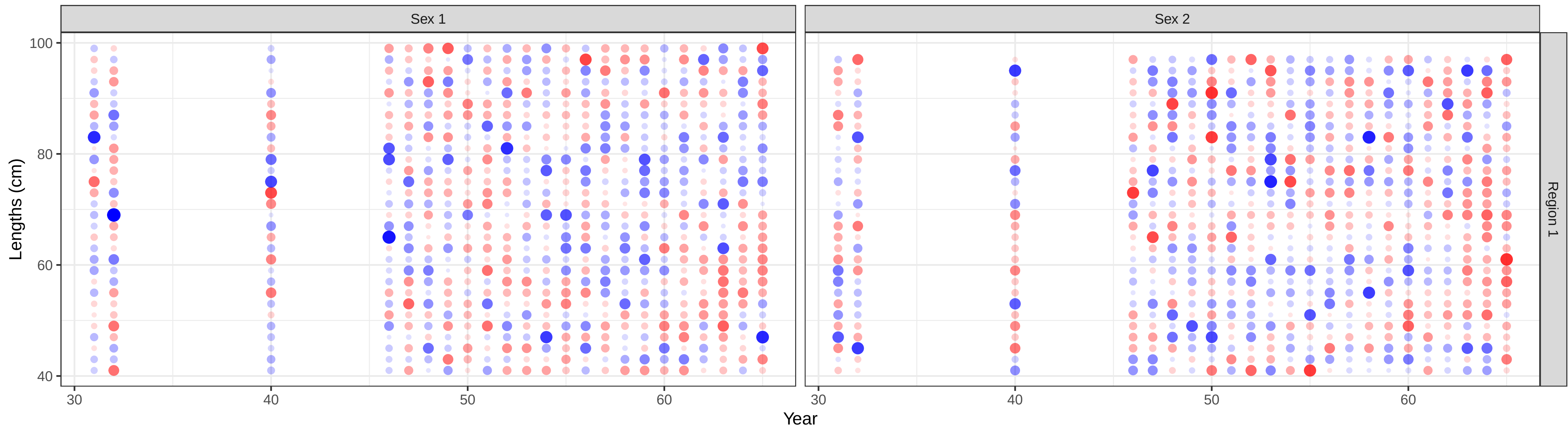


JPN LLS Length Comps Male

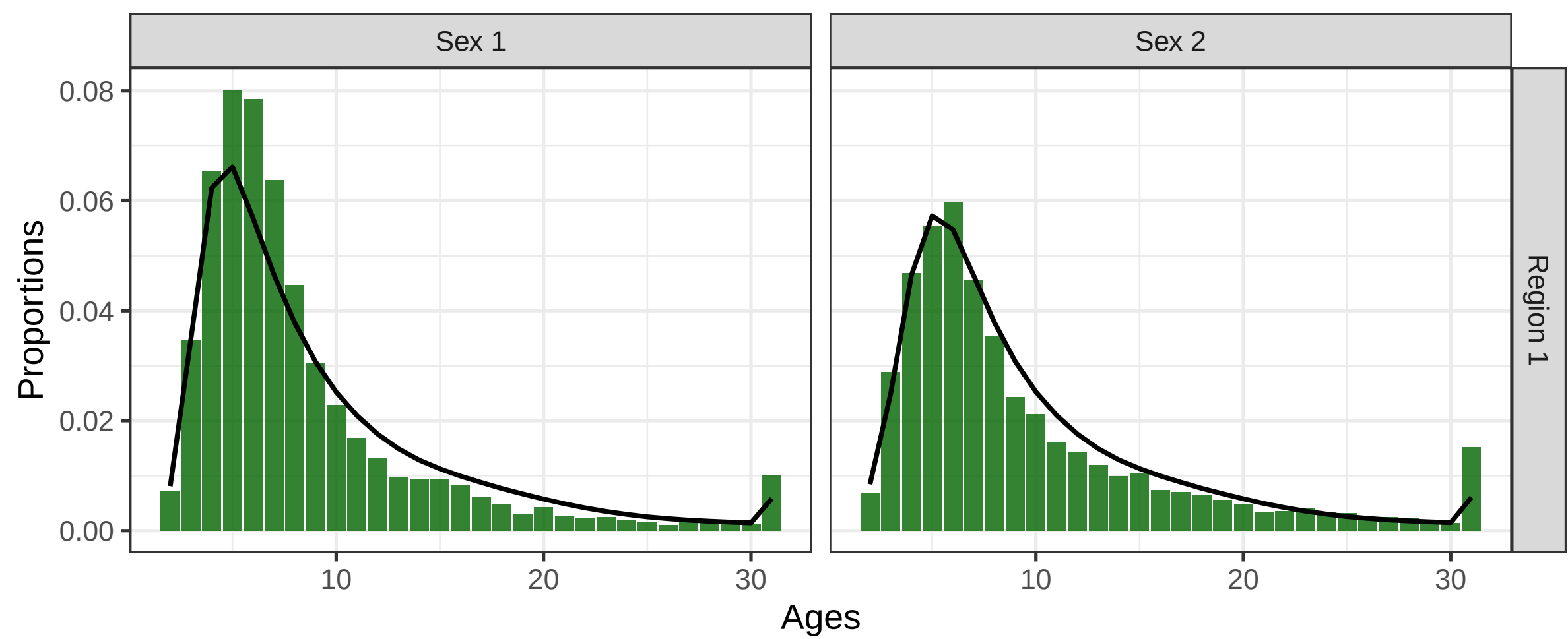
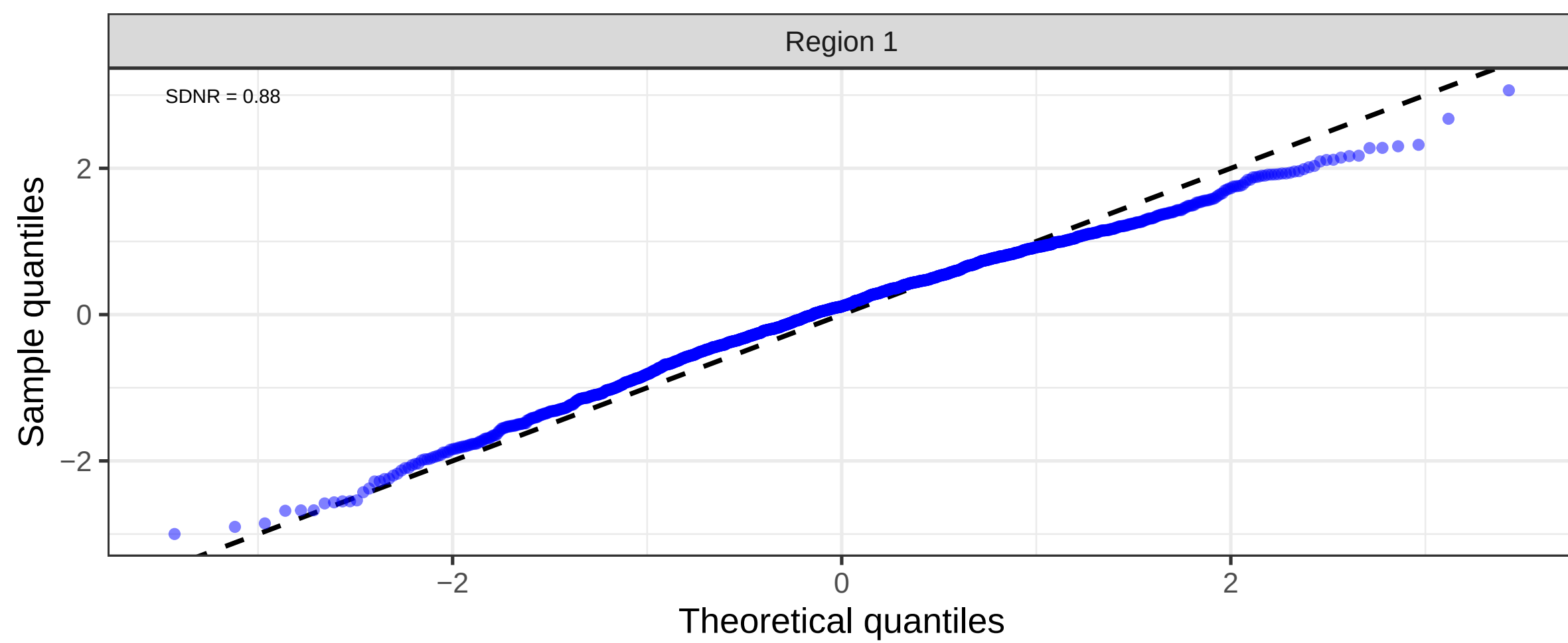
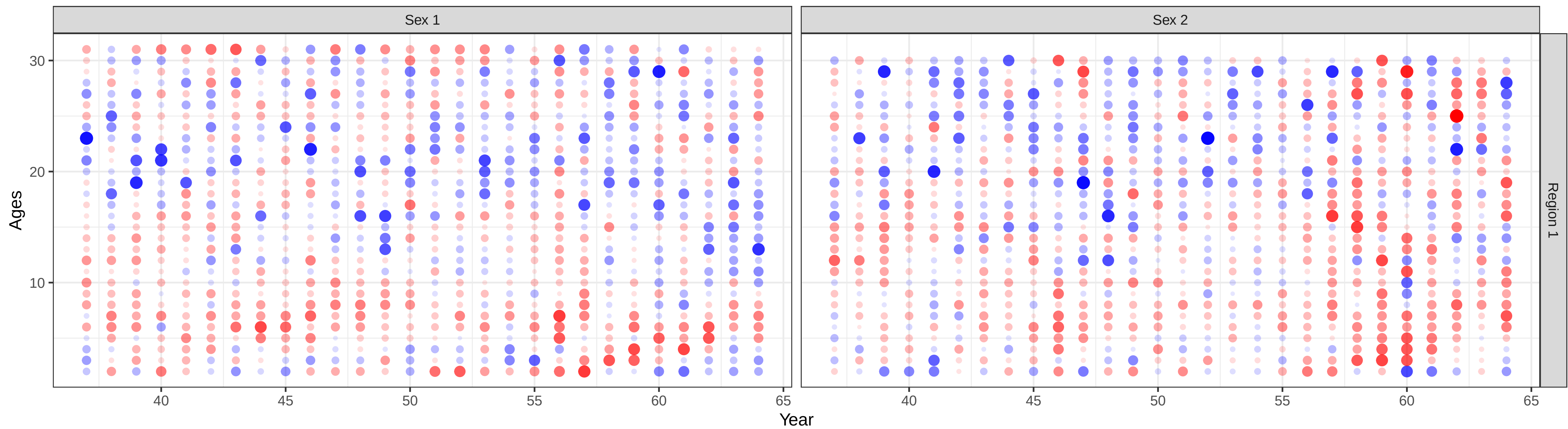


abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos

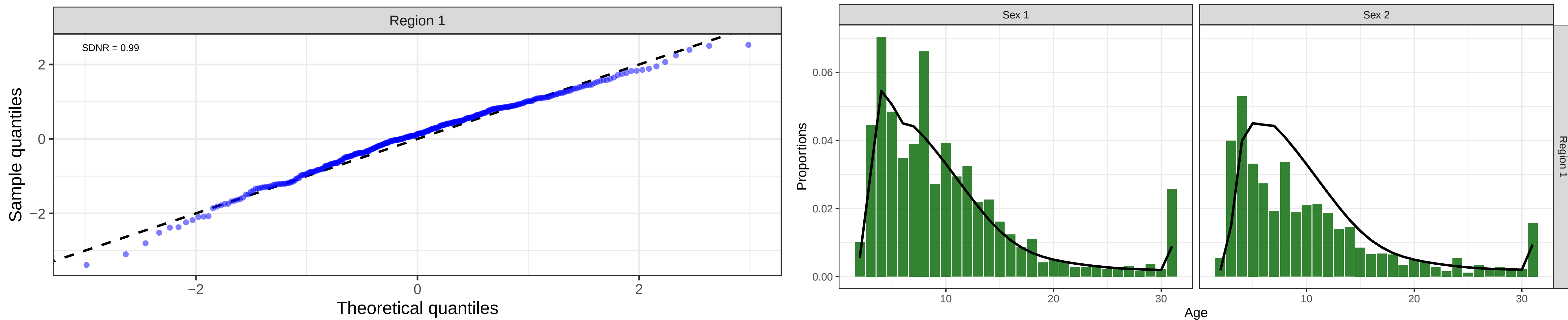
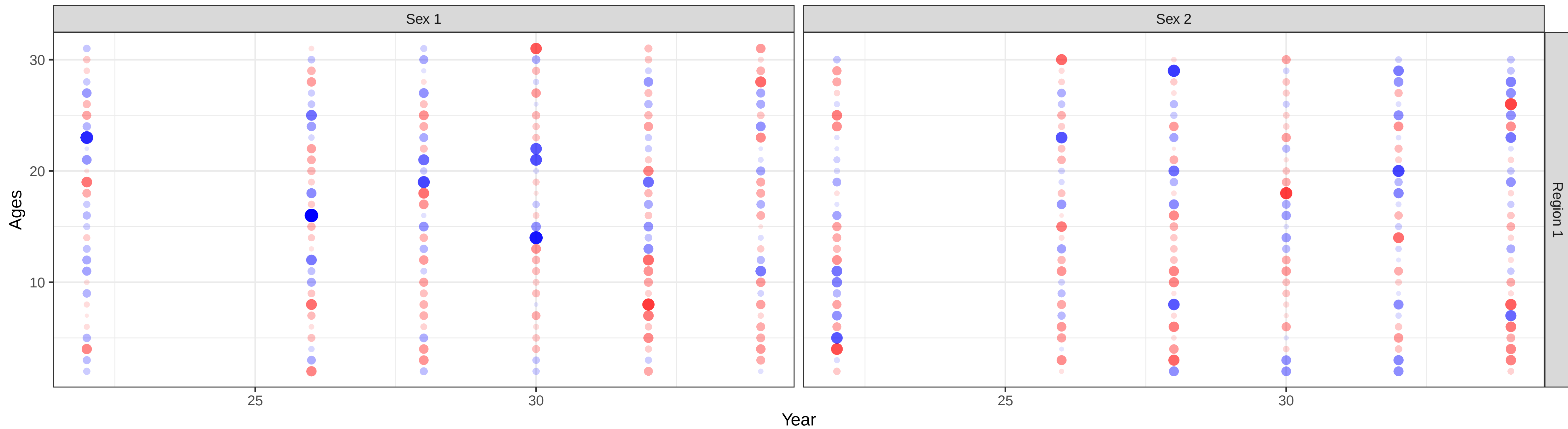




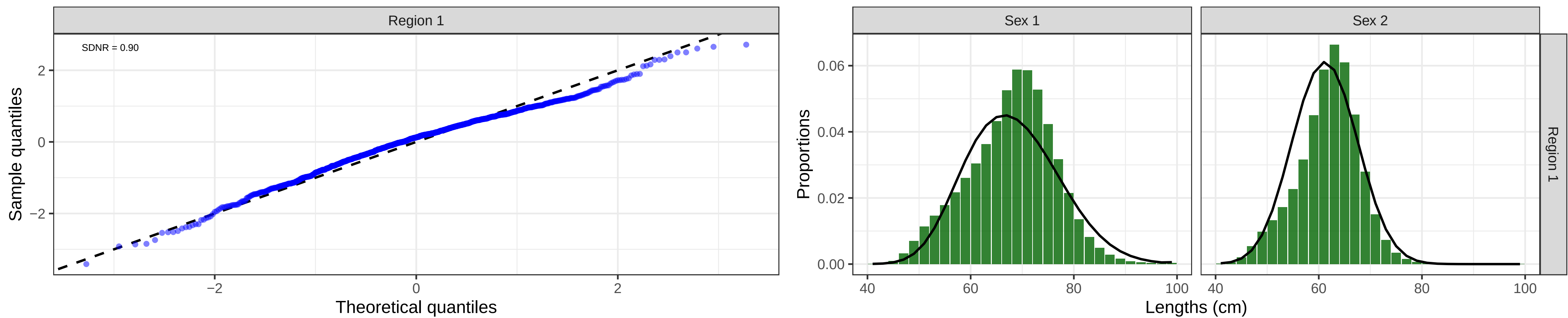
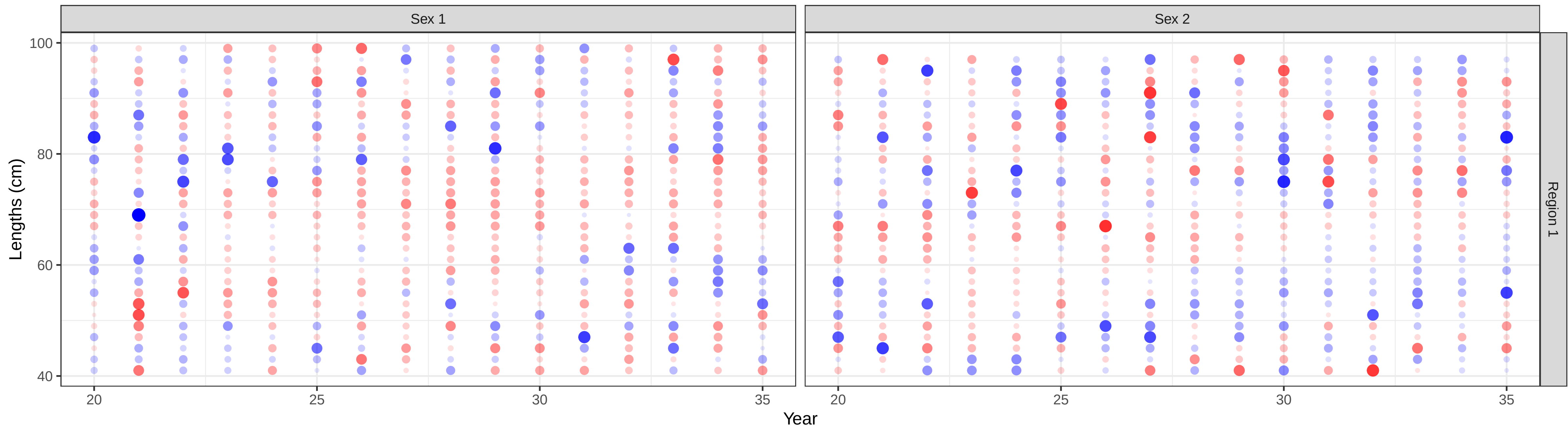
abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos



abs(Resid) 1 2 3 Sign ● Neg ● Pos

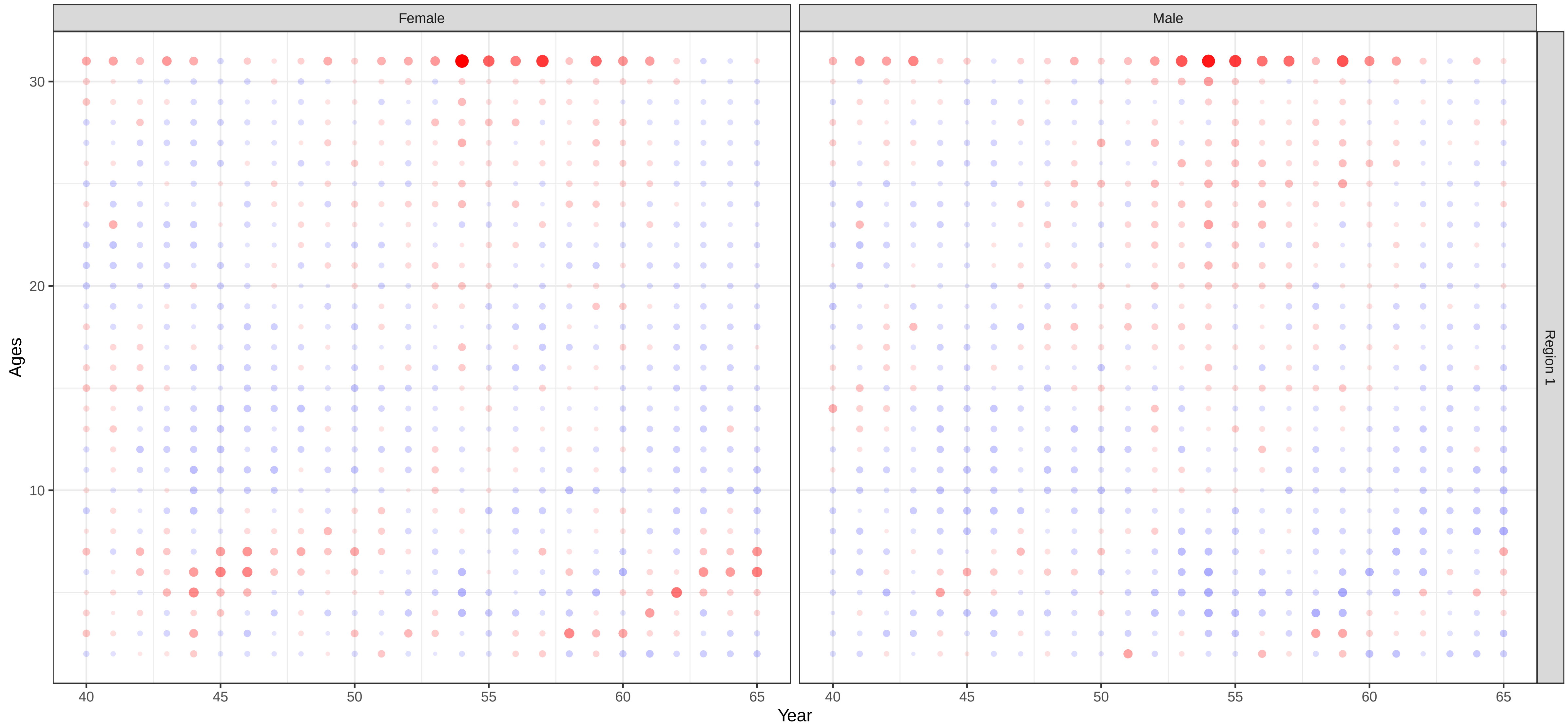


abs(Resid) ● 1 ● 2 ● 3 Sign ● Neg ● Pos



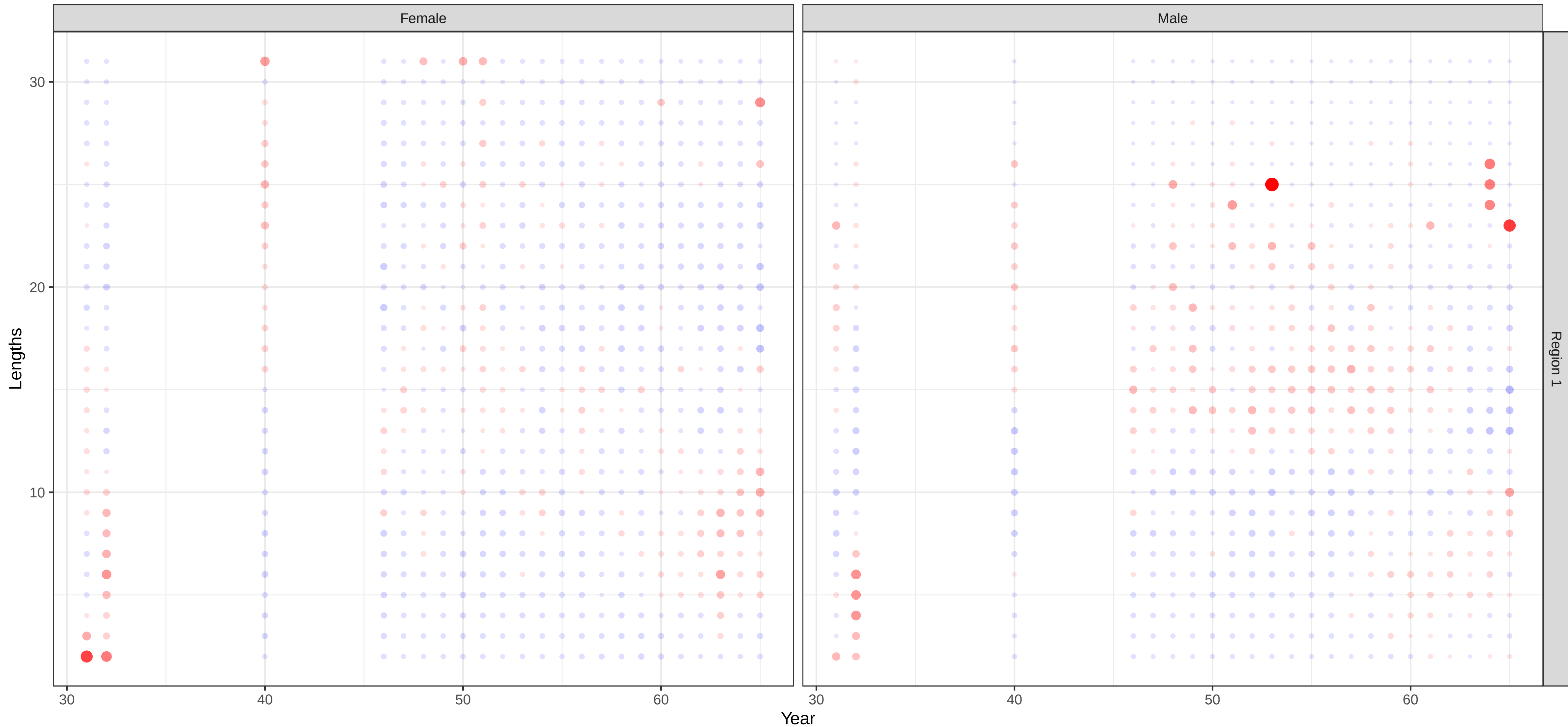
Fixed Gear Fishery Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0 ● 2.5



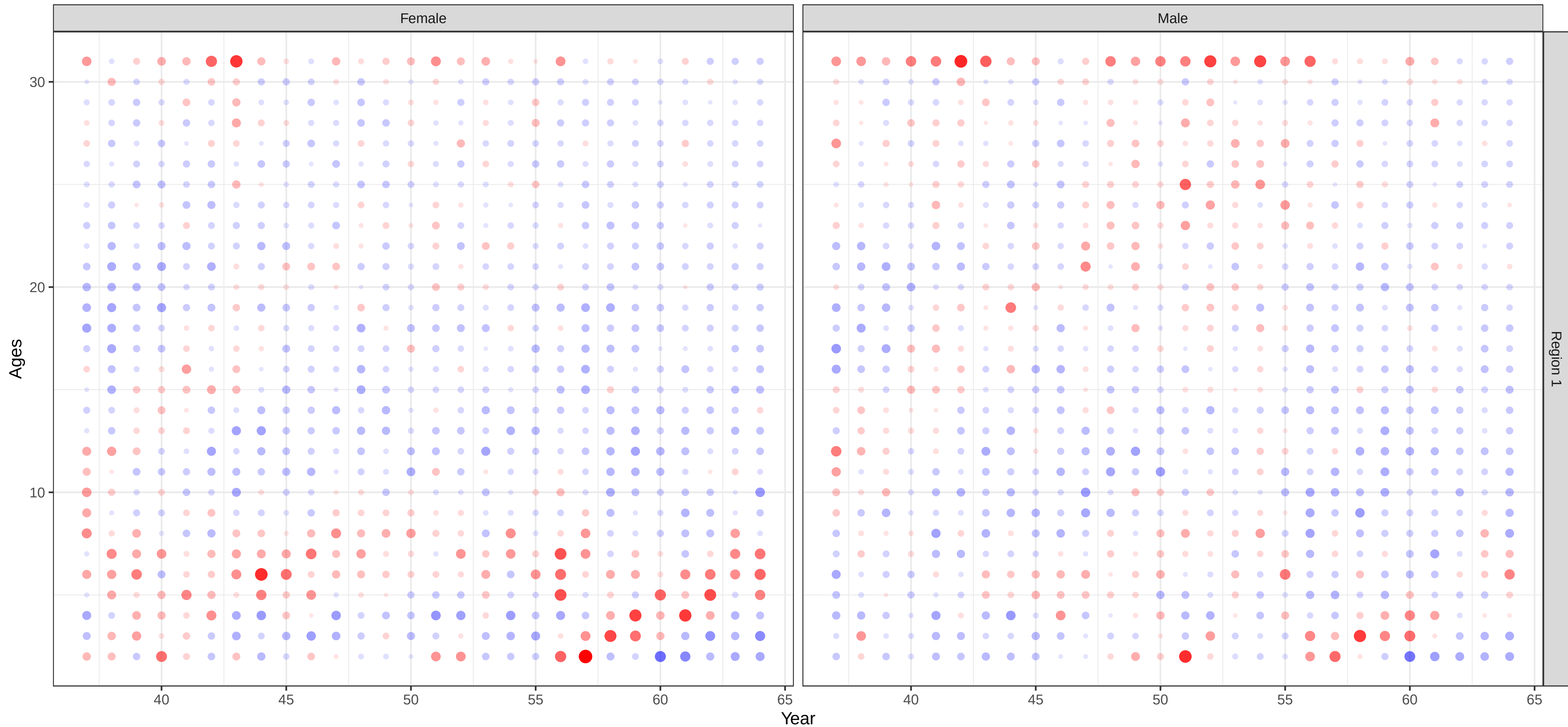
Trawl Fishery Length Compositions

Sign ● Neg ● Pos abs(Resid) ● 0 ● 1 ● 2 ● 3 ● 4



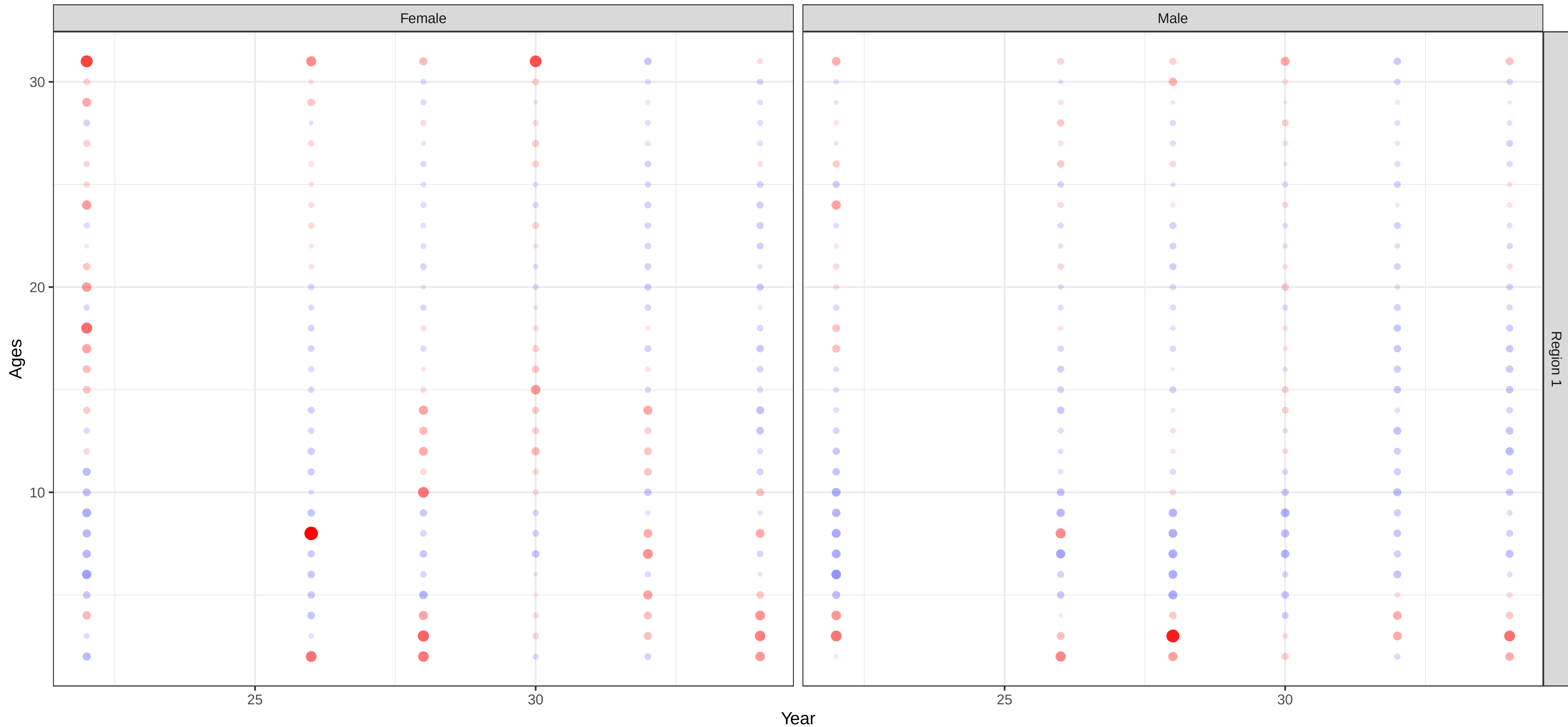
GOA LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0 ● 2.5



JPN LLS Age Compositions

Sign ● Neg ● Pos abs(Resid) ● 0.5 ● 1.0 ● 1.5 ● 2.0



JPN LLS Length Compositions

Sign ● Neg ● Pos

abs(Resid) ● 0.0 ● 1.0 ● 2.0
● 0.5 ● 1.5 ● 2.5

